

Traction Inverter

Chassis: Chassis2
BOM lines: 106 (98 priced)
Generated: 2026-07-17

Cost at quantity

Units	Estimated total
1	\$2,403.52
2	\$4,739.58
3	\$7,070.09
5	\$11,744.59
10	\$23,433.45

Contents

1. Order Summary
2. Fabrication Package
3. ControlBoard — schematic & PCB layers
4. DCBusCapacitorBoard — schematic & PCB layers
5. DCBusFilter — schematic & PCB layers
6. GateDriver — schematic & PCB layers
7. IOBoard — schematic & PCB layers
8. CapacitorTempSenseHarness — wiring harness
9. CurrentSenseHarness — wiring harness
10. GDVSHarness — wiring harness
11. TempSenseHarness — wiring harness

Order Summary

In-House Assembly

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-WH-TSBCM-A	CapacitorTempSenseHarness	-	N/A	N/A
4	HW-C2-WH-CS-A	CurrentSenseHarness	-	N/A	N/A
1	HW-C2-WH-GDVS-A	GDVSHarness	-	N/A	N/A
2	HW-C2-WH-TSIGBT-A	TempSenseHarness	-	N/A	N/A
				Subtotal	\$0.00

Digi-Key

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-HW-3M156058-ND -A	3M156058-ND	3M156058-ND	\$46.9800	\$46.98
				Subtotal	\$46.98

McMaster-Carr

Qty	Internal P/N	Description	Part Number	Unit	Total
8 (1pk)	HW-C2-MECH-90128A232 -A	90128A232	90128A232	\$10.5500/pk	\$10.55
5 (1pk)	HW-C2-MECH-90225A102 -A	90225A102	90225A102	\$12.8000/pk	\$12.80
18 (1pk)	HW-C2-MECH-91292A134 -A	91292A134	91292A134	\$11.7200/pk	\$11.72
4 (1pk)	HW-C2-MECH-91294A220 -A	91294A220	91294A220	\$12.7400/pk	\$12.74
6 (1pk)	HW-C2-MECH-91502A180 -A	91502A180	91502A180	\$19.6500/pk	\$19.65
6 (1pk)	HW-C2-MECH-91502A183 -A	91502A183	91502A183	\$17.2700/pk	\$17.27
6	HW-C2-MECH-94669A190 -A	94669A190	94669A190	\$1.3900	\$8.34
6	HW-C2-MECH-94669A196 -A	94669A196	94669A196	\$2.9600	\$17.76
6	HW-C2-MECH-94669A199 -A	94669A199	94669A199	\$3.1000	\$18.60
6	HW-C2-MECH-94669A202 -A	94669A202	94669A202	\$3.2200	\$19.32
25 (1pk)	HW-C2-MECH-98044A224 -A	98044A224	98044A224	\$14.8800/pk	\$14.88
				Subtotal	\$163.63

Mouser

Qty	Internal P/N	Description	Part Number	Unit	Total
8	HW-C2-HW-0022011042-A	0022011042	-	N/A	N/A
32	HW-C2-HW-08-50-0113-A	08-50-0113	-	N/A	N/A
70	HW-C2-CAP-100N-A	100nF 100V	80-C1210C104K1RAC	\$0.1610	\$11.27
27	HW-C2-RES-10K-A	10k 1210	603-AC1210JR-0710K L	\$0.0500	\$1.35
21	HW-C2-CAP-10N-A	10nF 500V	187-CL32B103KGFNN NE	\$0.1430	\$3.00
37	HW-C2-CAP-10U-A	10uF 25V	187-CL32A106KAULN NE	\$0.1340	\$4.96
4	HW-C2-RES-120-A	120 1210	603-AC1210FR-07120 RL	\$0.1000	\$0.40
17	HW-C2-RES-1K-A	1k 1210	603-RC1210FR-071KL	\$0.0490	\$0.83
8	HW-C2-CAP-1U-A	1uf 50V	187-CL32B105KBHNN NE	\$0.2400	\$1.92
2	HW-C2-CAP-2U2-A	2.2uf 1210 100v	80-C1210C225J1RAU TO	\$0.5200	\$1.04
12	HW-C2-RES-2R7-A	2.7 ohm 1210	667-ERJ-14BQF2R7U	\$0.2340	\$2.81
16	HW-C2-RES-20K-A	20k 1210 resistor	667-ERJ-P14F2002U	\$0.2450	\$3.92
4	HW-C2-HW-22-04-1041-A	22-04-1041	538-22-04-1041	\$0.8900	\$3.56
1	HW-C2-RES-220-A	220 1210	603-RC1210FR-07220 RL	\$0.1000	\$0.10
6	HW-C2-CAP-220P-A	220pF 100V C0G 1210	80-C1210C221J2G	\$0.8800	\$5.28
4	HW-C2-CAP-22P-A	22pF 50V	80-C1210C220J5G	\$0.7400	\$2.96
3	HW-C2-RES-3K-A	3k 1210 1%	603-RC1210FR-073KL	\$0.1000	\$0.30
3	HW-C2-RES-4K7-A	4.7k 1210	71-CRCW12104K70F KEAH	\$0.5900	\$1.77
3	HW-C2-HW-455-2266-ND-A	455-2266-ND	306-XHP-2	\$0.1000	\$0.30
25	HW-C2-HW-464762-E-A	464762-E	305-464762	\$0.3970	\$9.93
1	HW-C2-RES-499K-A	499kohm 1210 resistor	667-ERJ-P14F4993U	\$0.4400	\$0.44
5	HW-C2-HW-524265-E-A	524265-E	305-524265-E	\$7.8700	\$39.35
5	HW-C2-HW-544959-E-A	544959-E	-	N/A	N/A
2	HW-C2-RES-560-A	560ohm 1210 resistor	667-ERJ-P14F5600U	\$0.4400	\$0.88
1	HW-C2-HW-776231-1-A	776231-1	571-776231-1	\$16.6900	\$16.69
2	HW-C2-IC-8MHZ-A	8mhz hc49-4h crystal	710-830003156B	\$0.5000	\$1.00
8	HW-C2-HW-971150581-A	971150581	710-971150581	\$0.6000	\$4.80
8	HW-C2-HW-971250581-A	971250581	710-971250581	\$1.1000	\$8.80
8	HW-C2-HW-971350581-A	971350581	710-971350581	\$1.4800	\$11.84
8	HW-C2-HW-9774030592R-A	9774030592R	710-9774030592R	\$1.5700	\$12.56
2	HW-C2-HW-ACT45B-101-2P-TL003-A	ACT45B-101-2P-TL003	810-ACT45B1012PTL 003	\$1.5300	\$3.06
2	HW-C2-HW-AWHS105D00G-A	AWHS105D00G	523-AWHS105D00G	\$1.1100	\$2.22
3	HW-C2-HW-B2B-XH-AM_LF__SN__LF__SN__-A	B2B-XH-AM_LF__SN__LF__SN__-A	306-B2B-XH-AMLFSN P	\$0.1000	\$0.30

Qty	Internal P/N	Description	Part Number	Unit	Total
12	HW-C2-HW-B58031U9254 M062-A	B58031U9254M062	871-B58031U9254M062	\$4.6500	\$55.80
12	HW-C2-HW-BAT54WS-7-F-A	BAT54WS-7-F	621-BAT54WS-F	\$0.1240	\$1.49
5	HW-C2-HW-BTS462TATMA1-A	BTS462TATMA1	726-BTS462TATMA1	\$2.6600	\$13.30
3	HW-C2-HW-CM600DY-24T-A	CM600DY-24T	CM600DY-24T	\$323.5000	\$970.50
1	HW-C2-IC-CY15B102Q-SXET-A	CY15B102Q-SXET	727-CY15B102Q-SXET	\$18.1400	\$18.14
1	HW-C2-HW-DRDNB21D-7-A	DRDNB21D-7	621-DRDNB21D-7	\$0.5600	\$0.56
1	HW-C2-HW-EC7BW-110S12-A	EC7BW-110S12	418-EC7BW-110S12	\$48.8600	\$48.86
12	HW-C2-HW-FDFNYD1-110-5-A	FDFNYD1-110(5)	161-FDFNYD1-1105	\$0.4800	\$5.76
4	HW-C2-IC-ISO1042BDWVR-A	ISO1042BDWVR	595-ISO1042BDWVR	\$4.4400	\$17.76
1	HW-C2-IC-ISO1212DBQ-A	ISO1212DBQ	595-ISO1212DBQ	\$1.7300	\$1.73
4	HW-C2-HW-LA37S600S05KM-A	LA37S600S05KM	838-LA37S600S05KM	\$19.9600	\$79.84
1	HW-C2-HW-LT3080IQ-PBF-A	LT3080IQ#PBF	584-LT3080IQ#PBF	\$8.5400	\$8.54
2	HW-C2-HW-LTST-C230CKT-A	LTST-C230CKT	859-LTST-C230CKT	\$0.4100	\$0.82
2	HW-C2-HW-LTST-C230GKT-A	LTST-C230GKT	859-LTST-C230GKT	\$0.3600	\$0.72
1	HW-C2-IC-MAX22530AWE-A	MAX22530AWE+	700-MAX22530AWE+	\$15.6000	\$15.60
1	HW-C2-HW-MF-RHT200_32-2-A	MF-RHT200_32-2	652-MF-RHT200/32-2	\$0.6100	\$0.61
6	HW-C2-HW-MGJ2D121509MPC-R7-A	MGJ2D121509MPC-R7	580-MGJ2D121509MPC-R7	\$6.7700	\$40.62
6	HW-C2-HW-MKP1848S61010JY5B-A	MKP1848S61010JY5B	75-MKP1848S61010JY5B	\$7.5400	\$45.24
5	HW-C2-HW-MM3Z3V0ST1G-A	MM3Z3V0ST1G	863-MM3Z3V0ST1G	\$0.1700	\$0.85
6	HW-C2-IC-NCV57100DWR2G-A	NCV57100DWR2G	863-NCV57100DWR2G	\$2.9800	\$17.88
1	HW-C2-HW-NXFT15XH103FEAB021-A	NXFT15XH103FEAB021	81-NXFT15XH103FEAB021	\$0.3400	\$0.34
1	HW-C2-HW-RD7-12S033R-A	RD7-12S033R	737-RD7-12S033R	\$6.7700	\$6.77
3	HW-C2-HW-RKE-1205S_H-A	RKE-1205S_H	919-RKE-1205S/H	\$5.1700	\$15.51
2	HW-C2-HW-RTS103C1R2M6L201-A	RTS103C1R2M6L201	527-RTS103C1R2M6L201	\$2.2700	\$4.54
2	HW-C2-HW-S3N-A	S3N	512-S3N	\$0.6700	\$1.34
2	HW-C2-HW-SSW-103-01-L-D-A	SSW-103-01-L-D	200-SSW10301LD	\$1.3400	\$2.68
1	HW-C2-HW-SSW-104-01-L-D-A	SSW-104-01-L-D	200-SSW10401LD	\$1.6900	\$1.69

Qty	Internal P/N	Description	Part Number	Unit	Total
2	HW-C2-HW-SSW-108-01-L-D-A	SSW-108-01-L-D	200-SSW10801LD	\$2.3500	\$4.70
1	HW-C2-CONN-SSW-110-01-F-D-A	SSW-110-01-F-D	200-SSW11001FD	\$2.4100	\$2.41
1	HW-C2-HW-SSW-110-01-F-D-A	SSW-110-01-F-D	-	N/A	N/A
1	HW-C2-IC-STM32G474RCT3-A	STM32G474RCT3	511-STM32G474RCT3	\$8.3100	\$8.31
1	HW-C2-IC-STM32H723ZGT6-A	STM32H723ZGT6	511-STM32H723ZGT6	\$13.2500	\$13.25
1	HW-C2-HW-STS41A-AWLB-R3-A	STS41A-AWLB-R3	403-STS41A-AWLB-R3	\$1.1300	\$1.13
6	HW-C2-HW-STTH112A-A	STTH112A	511-STTH112A	\$0.5900	\$3.54
6	HW-C2-HW-SXH-001T-P0.6-A	SXH-001T-P0.6	306-SXH-001T-P0.6	\$0.1000	\$0.60
4	HW-C2-HW-SZMMSZ4678T1G-A	SZMMSZ4678T1G	863-SZMMSZ4678T1G	\$0.4300	\$1.72
21	HW-C2-HW-TESTPOINT-TH-RED-A	Testpoint TH, RED	534-5000	\$0.2930	\$6.15
16	HW-C2-RES-250K-A	Thick Film Resistors - SMD 1/2watt 249Kohms 1%	71-CRCW1210-249K-E3	\$0.0700	\$1.12
1	HW-C2-HW-TPS389006ADJRTER-A	TPS389006ADJRTER	595-TPS389006ADJRTER	\$5.1000	\$5.10
2	HW-C2-HW-TS40-66-50-R-260-SMT-TR-A	TS40-66-50-R-260-SMT-TR	179-TS406650R260S	\$0.4800	\$0.96
2	HW-C2-HW-TSW-103-18-G-D-A	TSW-103-18-G-D	200-TSW10318GD	\$1.0100	\$2.02
2	HW-C2-HW-TSW-108-18-F-D-A	TSW-108-18-F-D	200-TSW10818FD	\$1.4500	\$2.90
1	HW-C2-HW-TSW-110-18-T-D-A	TSW-110-18-T-D	200-TSW11018TD	\$1.5800	\$1.58
1	HW-C2-CONN-TSW-110-21-L-D-LL-A	TSW-110-21-L-D-LL	200-TSW11021LDLL	\$2.2700	\$2.27
13	HW-C2-HW-UCM1H101MCL1GS-A	UCM1H101MCL1GS	647-UCM1H101MCL1GS	\$0.3250	\$4.23
60	HW-C2-HW-UCS2D331MHD-A	UCS2D331MHD	647-UCS2D331MHD	\$2.1600	\$129.60
1	HW-C2-HW-UJ2-B-HR-G-V-A	UJ2-B-HR-G-V	179-UJ2-B-HR-G-V	\$1.7000	\$1.70
18	HW-C2-HW-VY1222M47Y5UQ6TV0-A	VY1222M47Y5UQ6TV0	72-VY1222M47Y5UQ6TV0	\$0.2710	\$4.88
				Subtotal	\$1,733.29

PCB Fabrication

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-PCB-CTRL-A	ControlBoard	ControlBoard	\$25.2400	\$25.24
1	HW-C2-PCB-DCBUSCAP-A	DCBusCapacitorBoard	DCBusCapacitorBoard	\$7.5400	\$7.54
1	HW-C2-PCB-DCFLT-A	DCBusFilter	DCBusFilter	\$8.4400	\$8.44
1	HW-C2-PCB-GD-A	GateDriver	GateDriver	\$11.6600	\$11.66

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-PCB-IO-A	IOBoard	IOBoard	\$11.6600	\$11.66
				Subtotal	\$64.54

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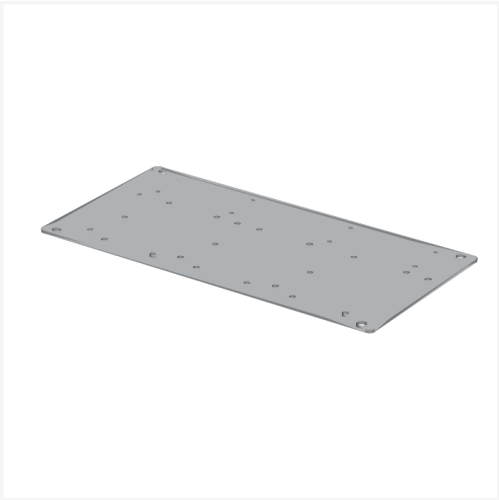
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1	HW-C2-PLT-BSP-A	HW-C2-BSP-A	HW-C2-BSP-A	\$201.6100	\$201.61
1	HW-C2-PLT-CHSP-A	HW-C2-CHSP-A	HW-C2-CHSP-A	\$33.3100	\$33.31
2	HW-C2-BB-DCLBB-A	HW-C2-DCLBB-A	HW-C2-DCLBB-A	\$50.7100	\$101.42
3	HW-C2-BB-PBB-A	HW-C2-PBB-A	HW-C2-PBB-A	\$19.5800	\$58.74
				Subtotal	\$395.08

Fabrication Package

Item	Kind	Qty	Rev	Price	Ready
ControlBoard	PCB	1	A	\$25.24	yes
DCBusCapacitorBoard	PCB	1	A	\$7.54	yes
DCBusFilter	PCB	1	A	\$8.44	yes
GateDriver	PCB	1	A	\$11.66	yes
IOBoard	PCB	1	A	\$11.66	yes
HW-C2-BSP-A	fab part	1	A	\$201.61	yes
HW-C2-CHSP-A	fab part	1	A	\$33.31	yes
HW-C2-DCLBB-A	fab part	2	A	\$50.71	yes
HW-C2-PBB-A	fab part	3	A	\$19.58	yes
CapacitorTempSenseHarness	harness	-	-	-	yes
CurrentSenseHarness	harness	-	-	-	yes
GDVSHarness	harness	-	-	-	yes
TempSenseHarness	harness	-	-	-	yes

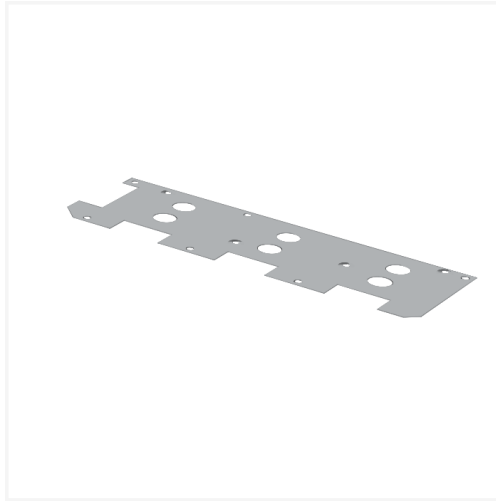
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6061 T6 Aluminum / 9.52 mm / Sheet Cutting / Deburring, Tapping



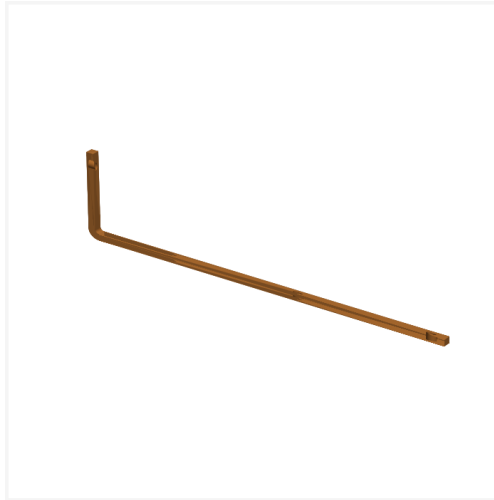
HW-C2-CHSP-A

6061 T6 Aluminum / 3.17 mm / Sheet Cutting / Deburring



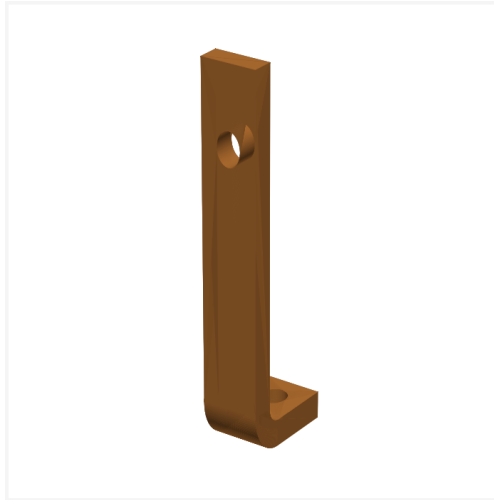
HW-C2-DCLBB-A

Copper / 4.75 mm / Sheet Cutting / Bending



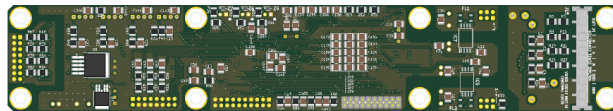
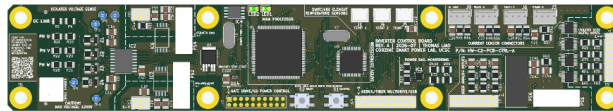
HW-C2-PBB-A

Copper / 4.75 mm / Sheet Cutting / Bending



CONTROLBOARD

HW-C2-PCB-CTRL-A · printed circuit board — schematic and PCB layers

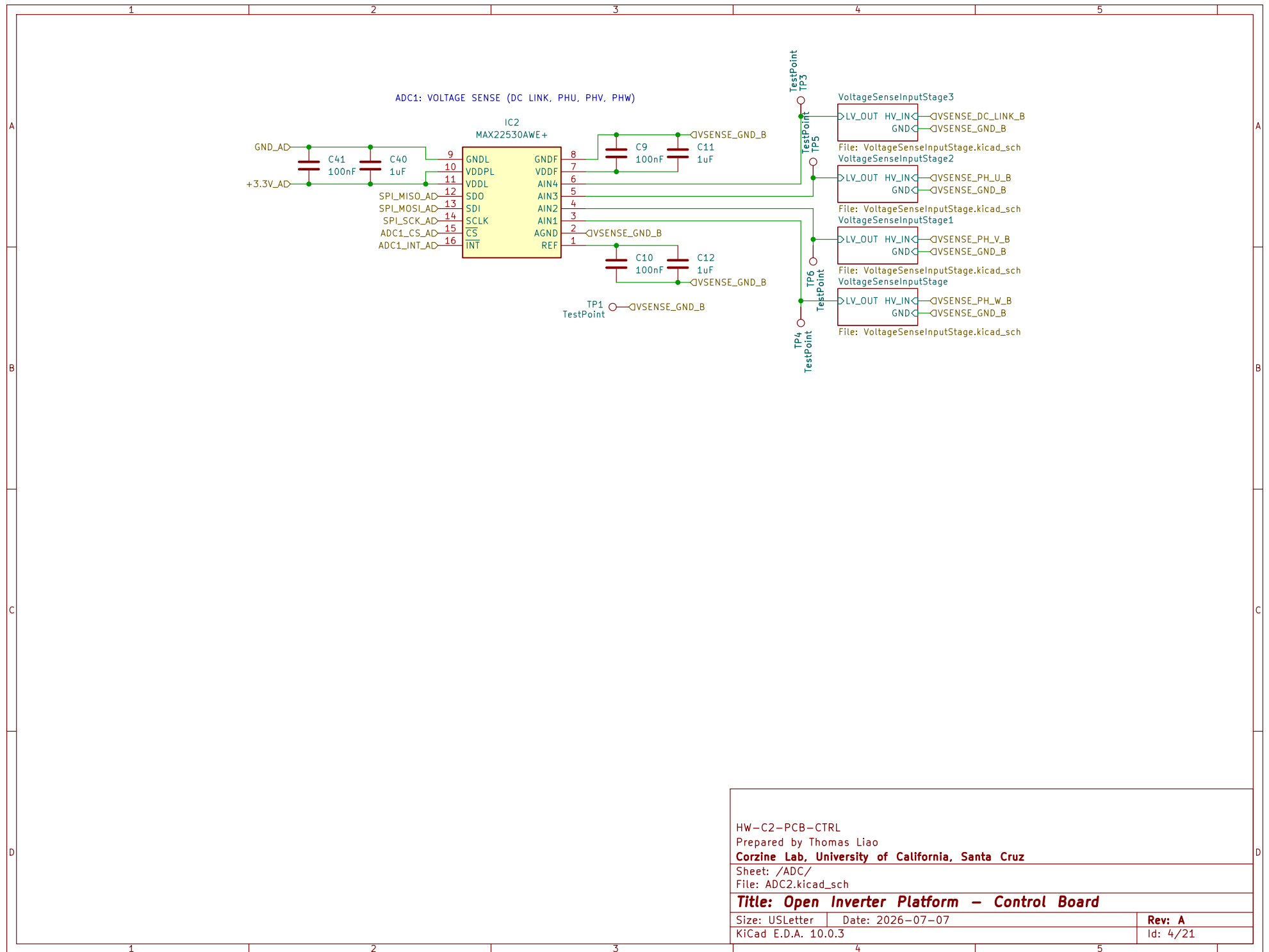


NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION,
DISABLE TIM1 SYNC PIN (TIM1 ETR)



Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3		Id: 1/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

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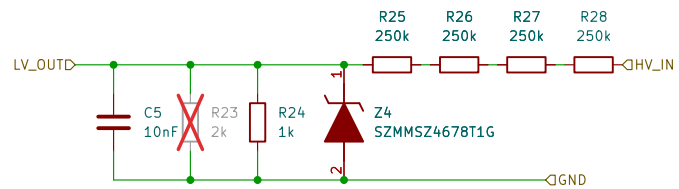
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Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 4/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

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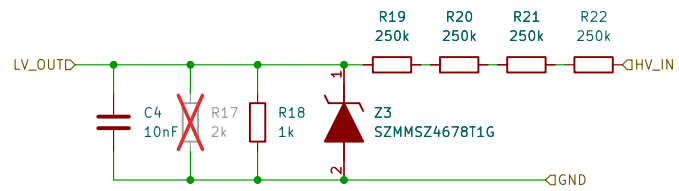
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Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 2/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

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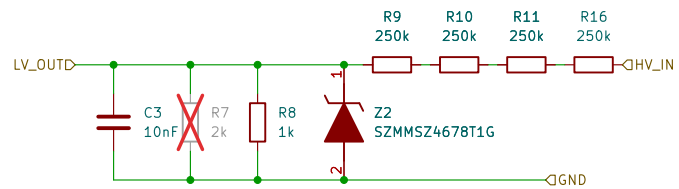
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Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 3/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

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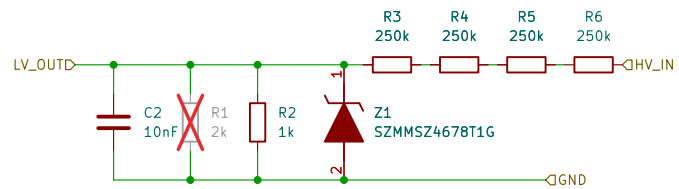
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Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 5/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/

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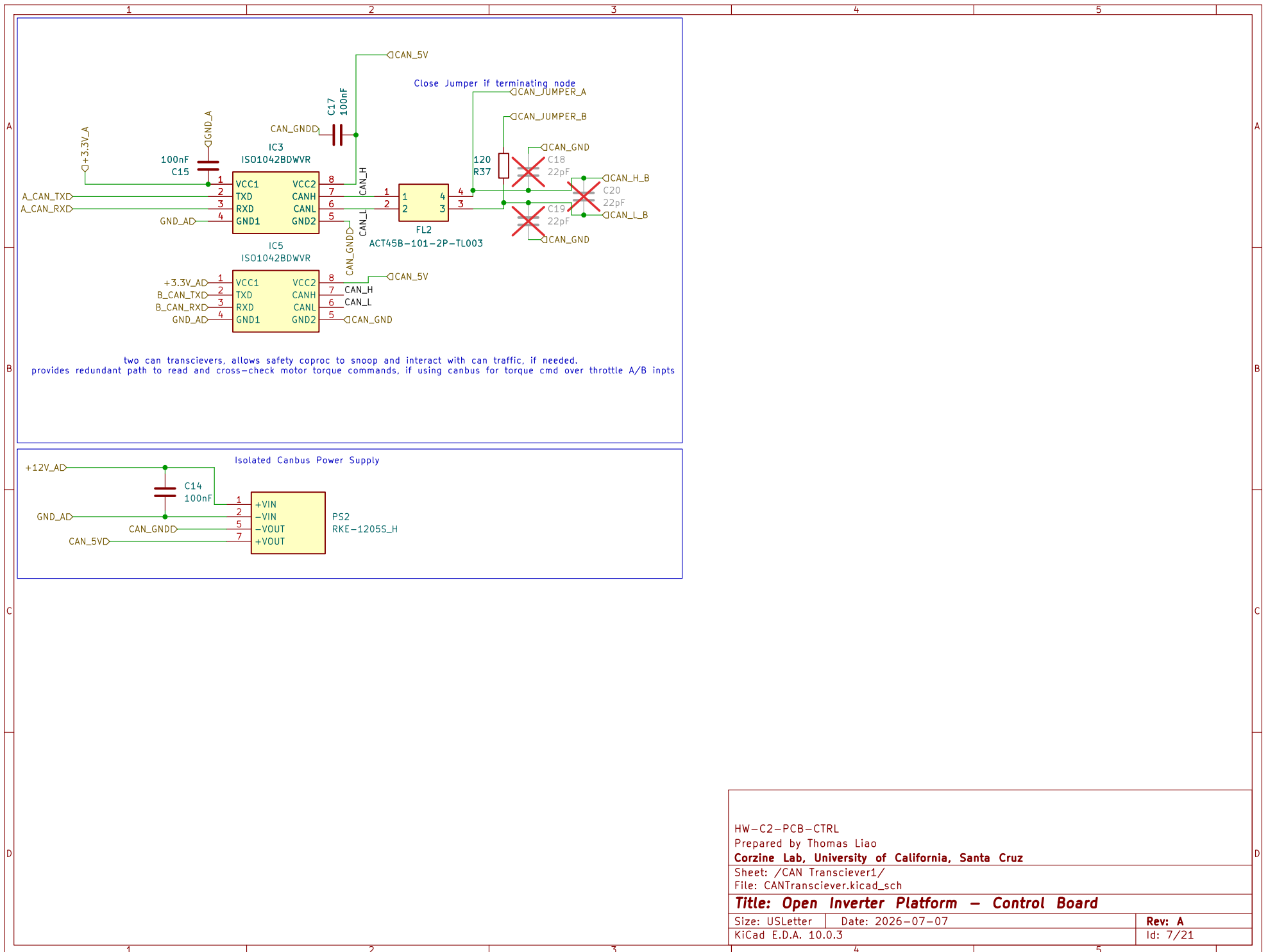
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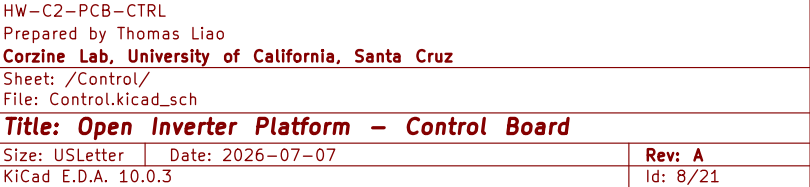
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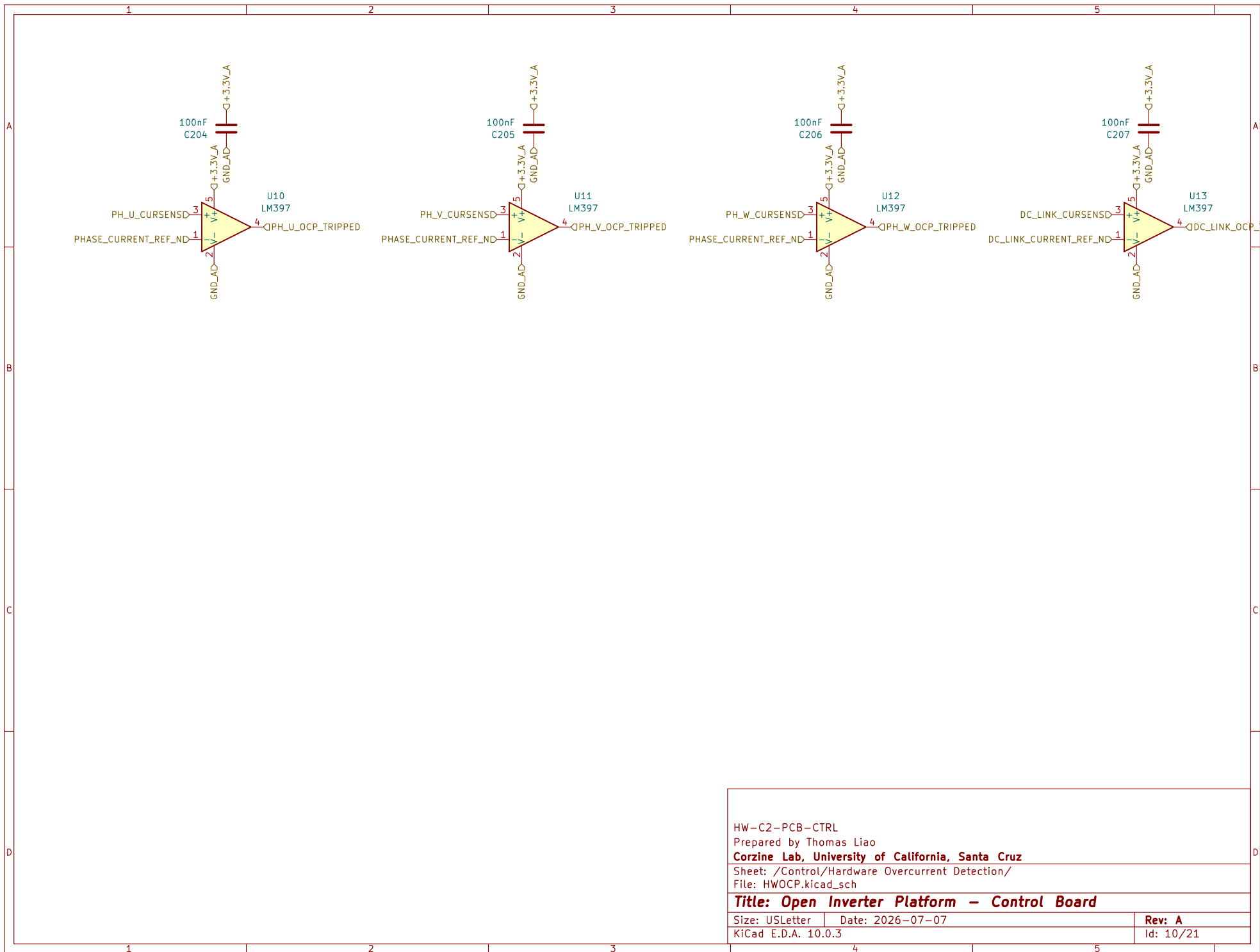
KiCad E.D.A. 10.0.3

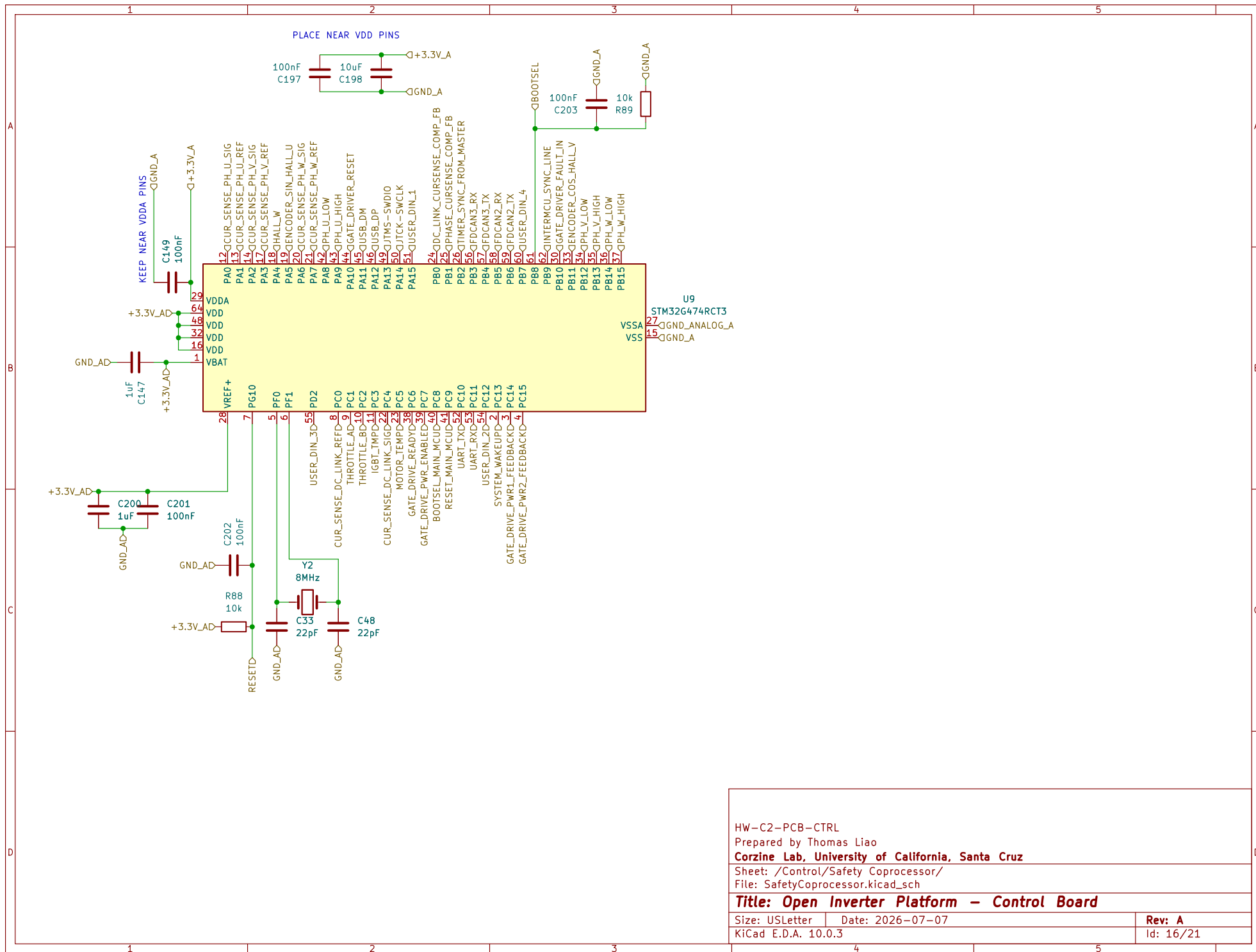
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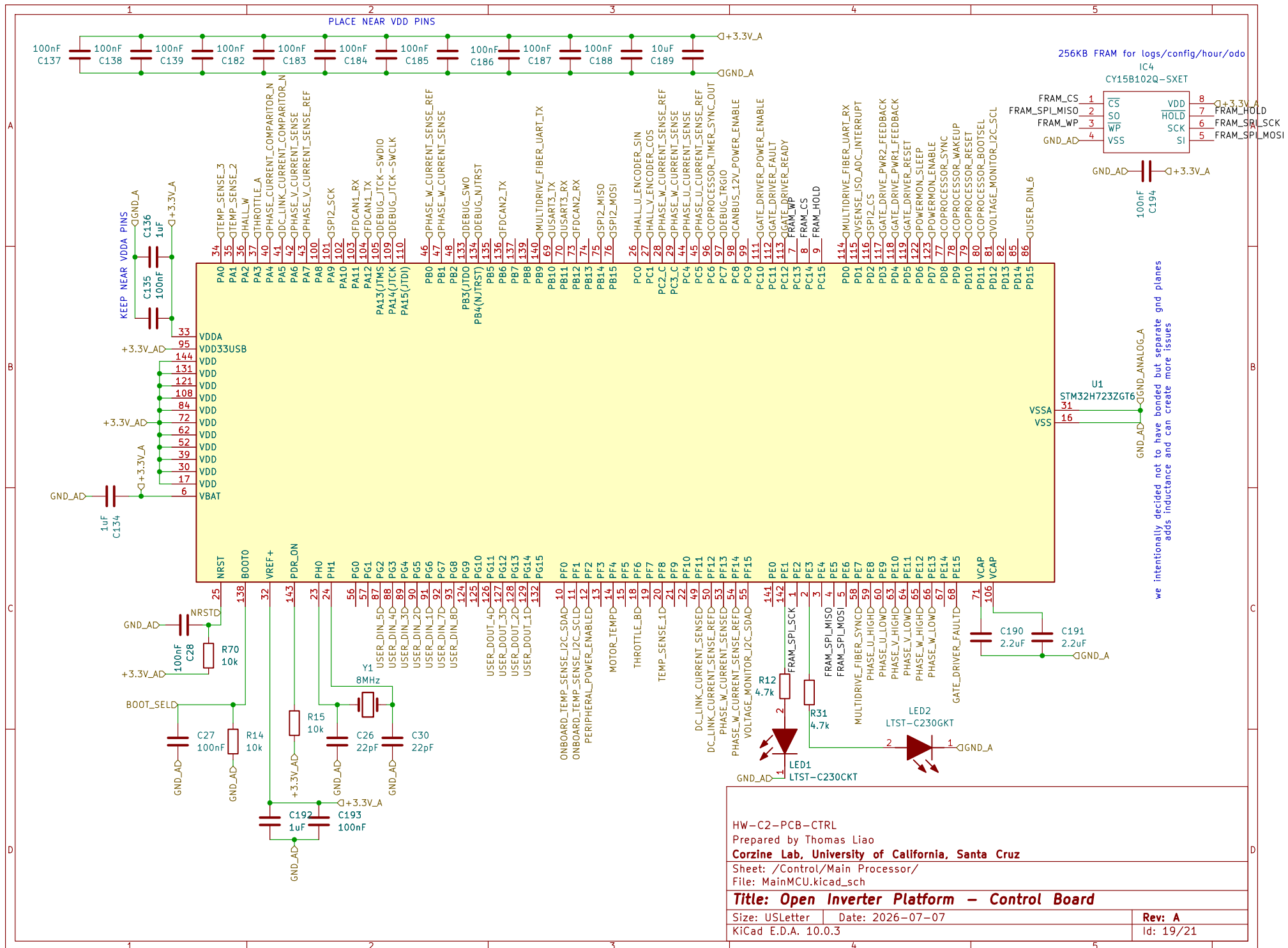
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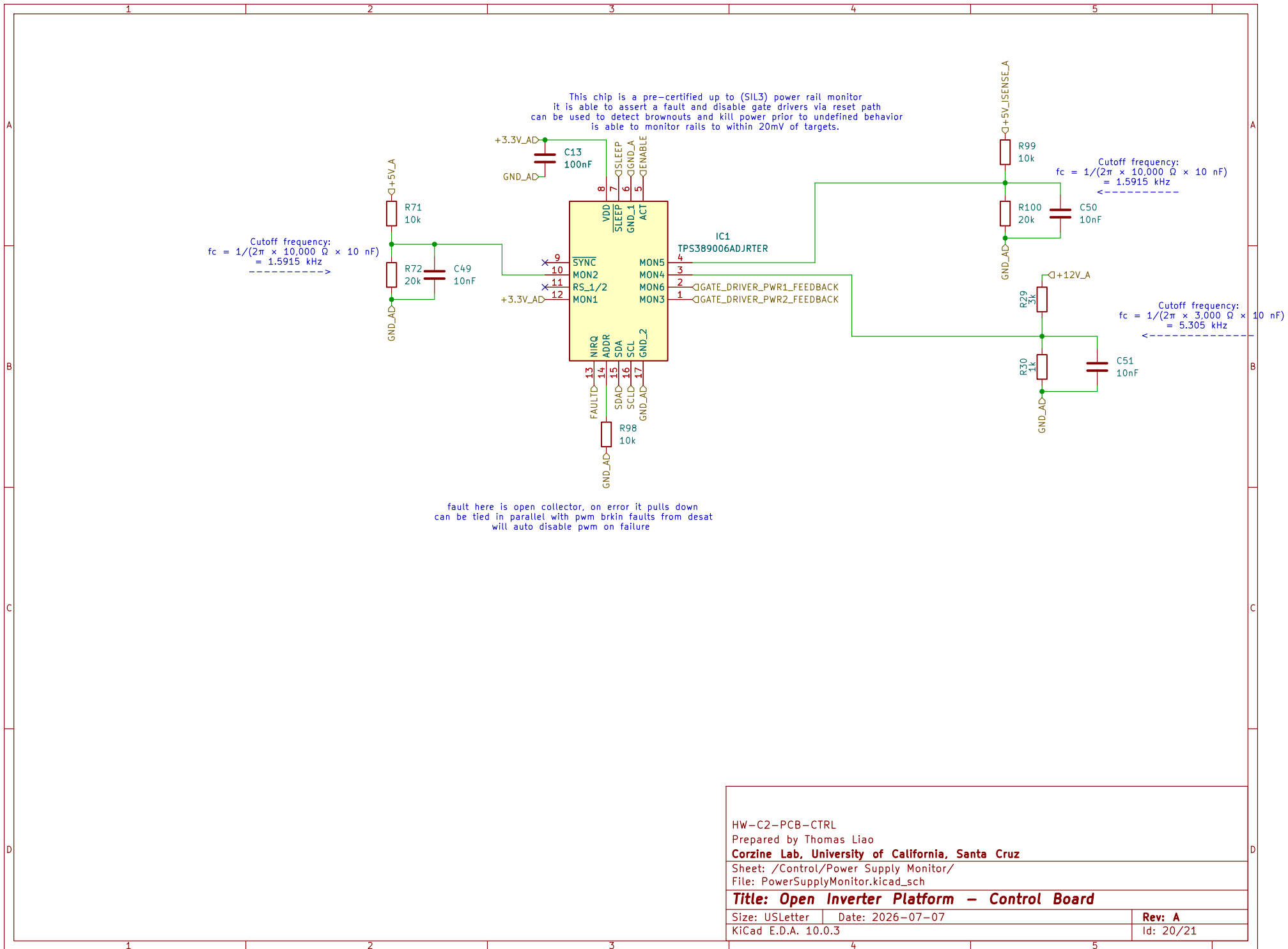


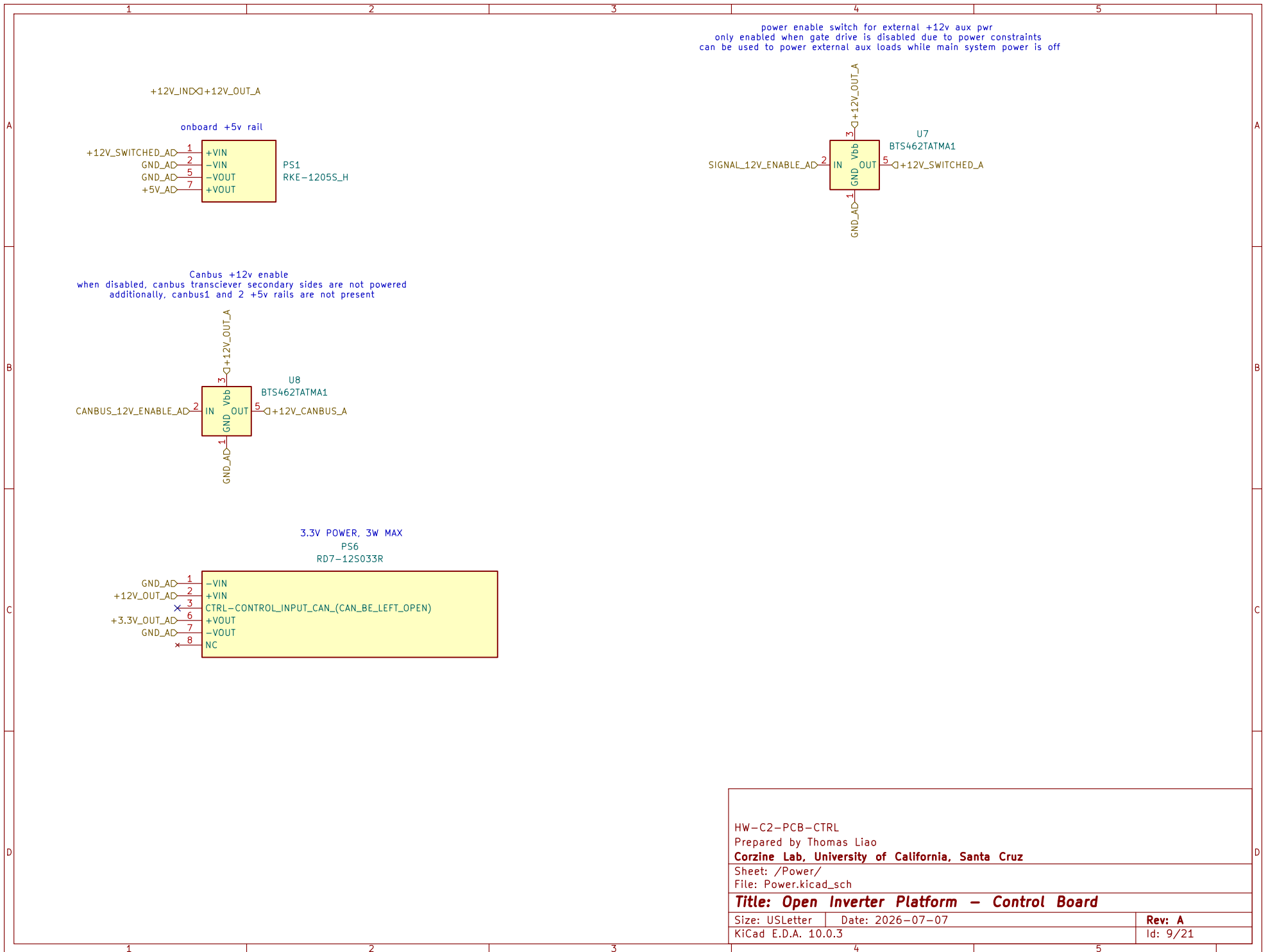


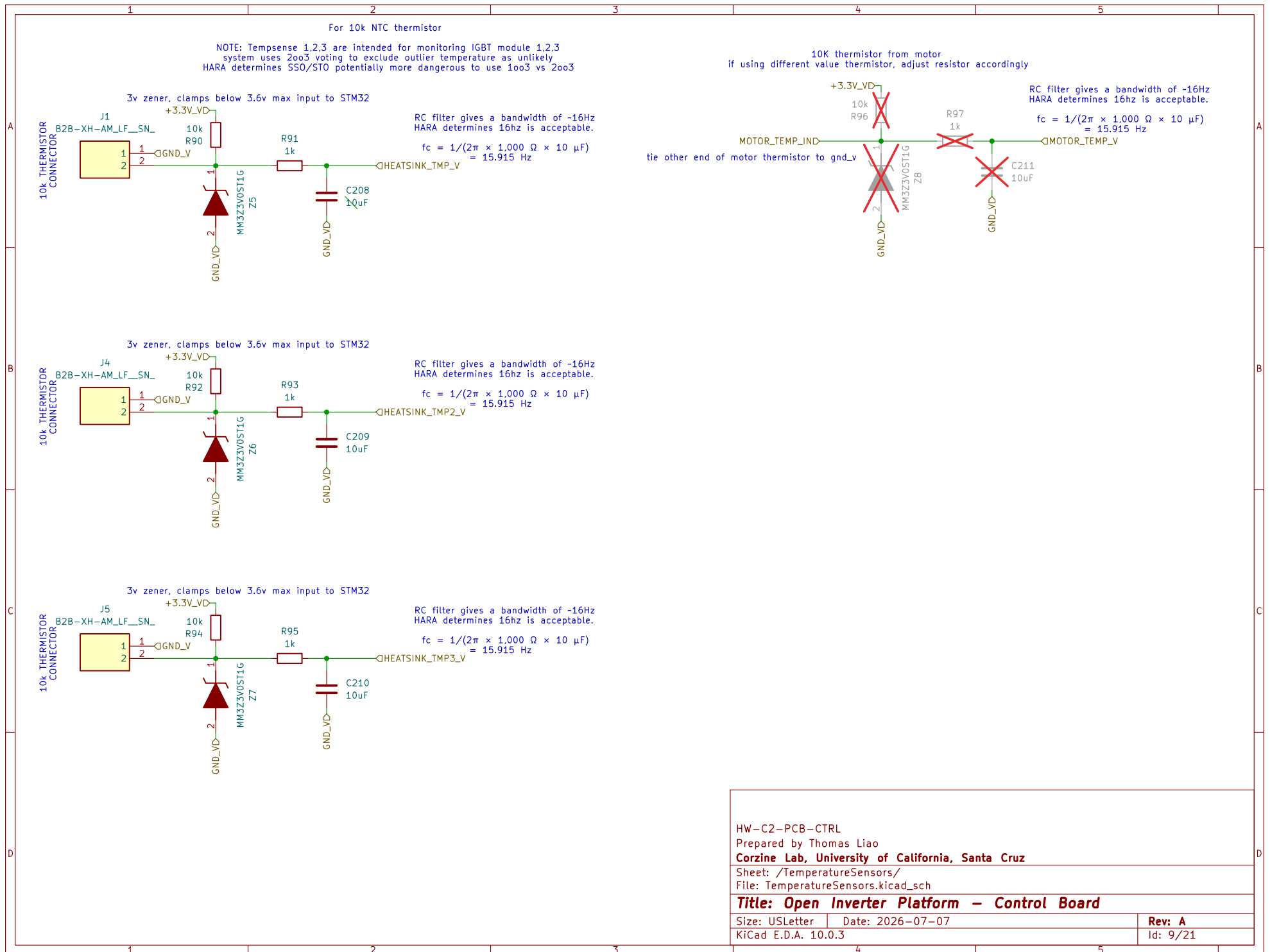












HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /TemperatureSensors/

File: TemperatureSensors.kicad_sch

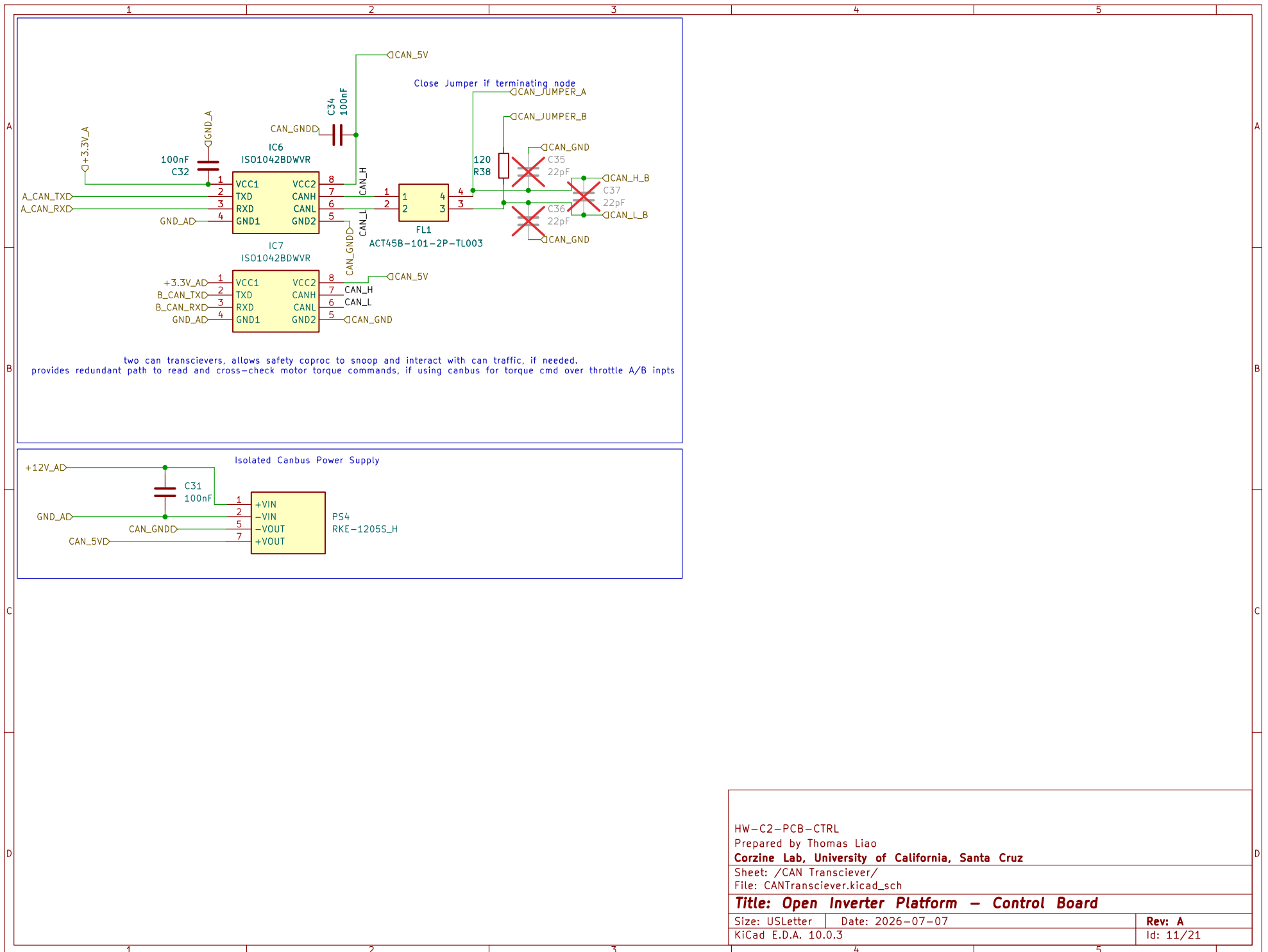
Title: Open Inverter Platform – Control Board

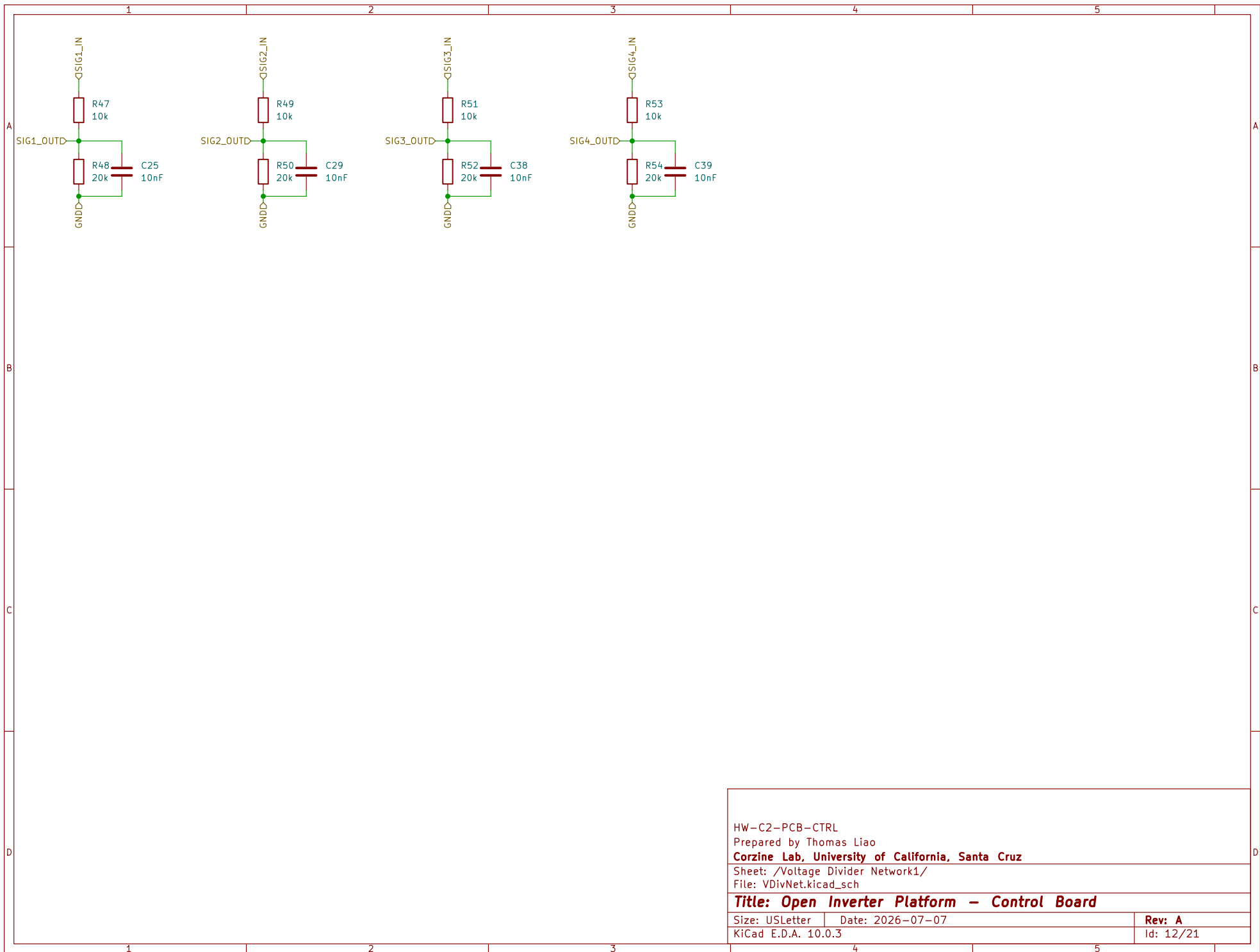
Size: USLetter Date: 2026-07-07

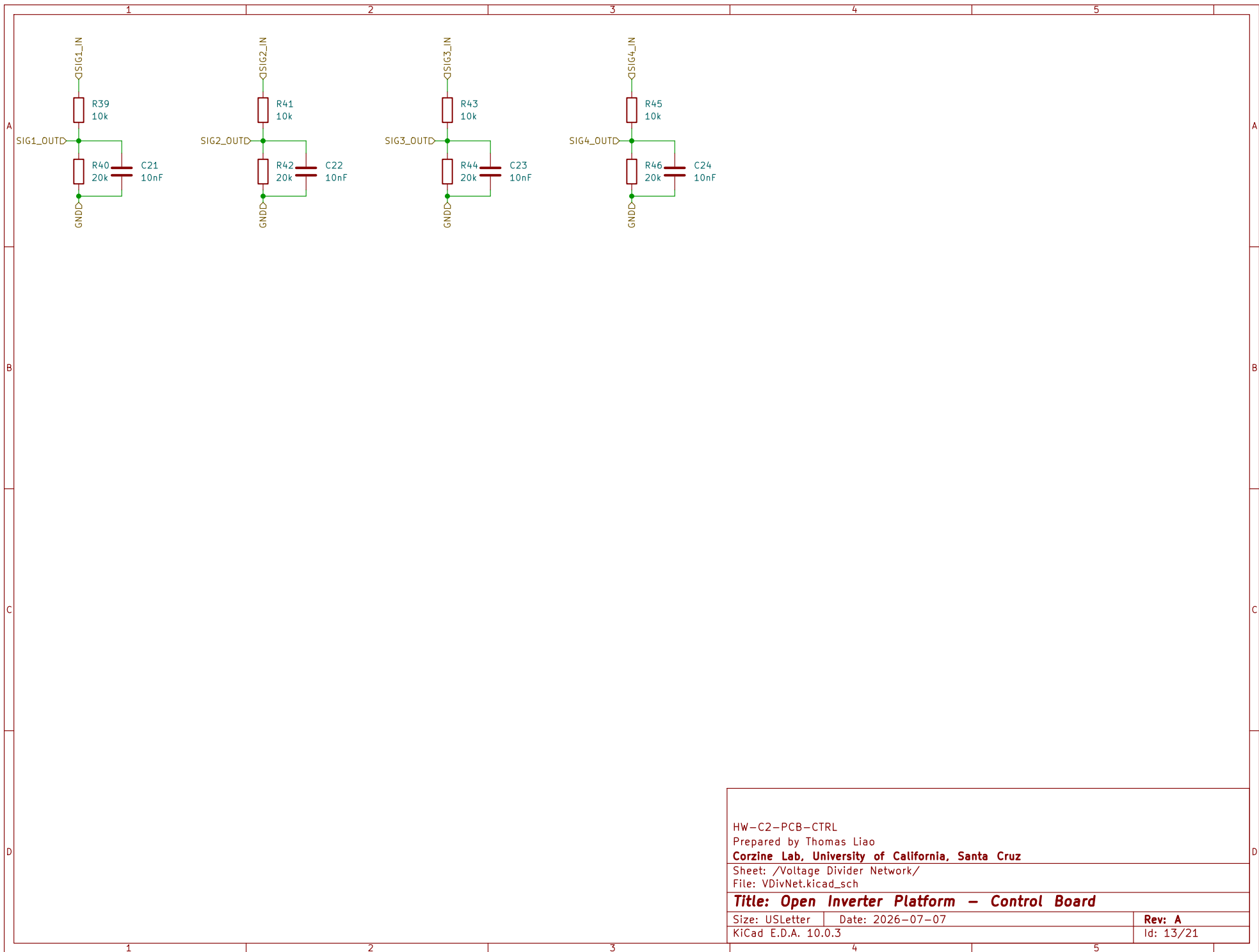
KiCad E.D.A. 10.0.3

Rev: A

Id: 9/21







HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network/

File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

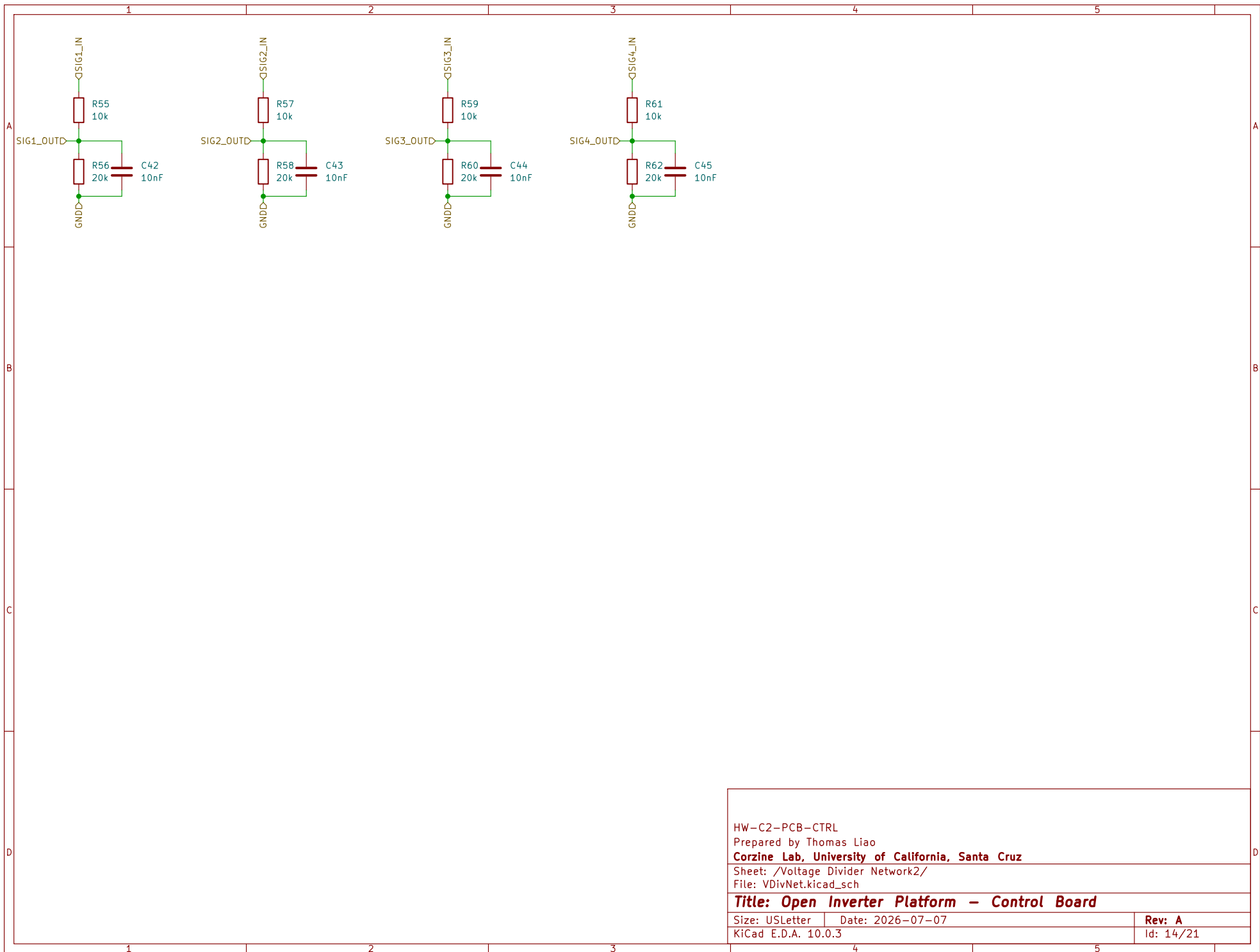
Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 13/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network2/

File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

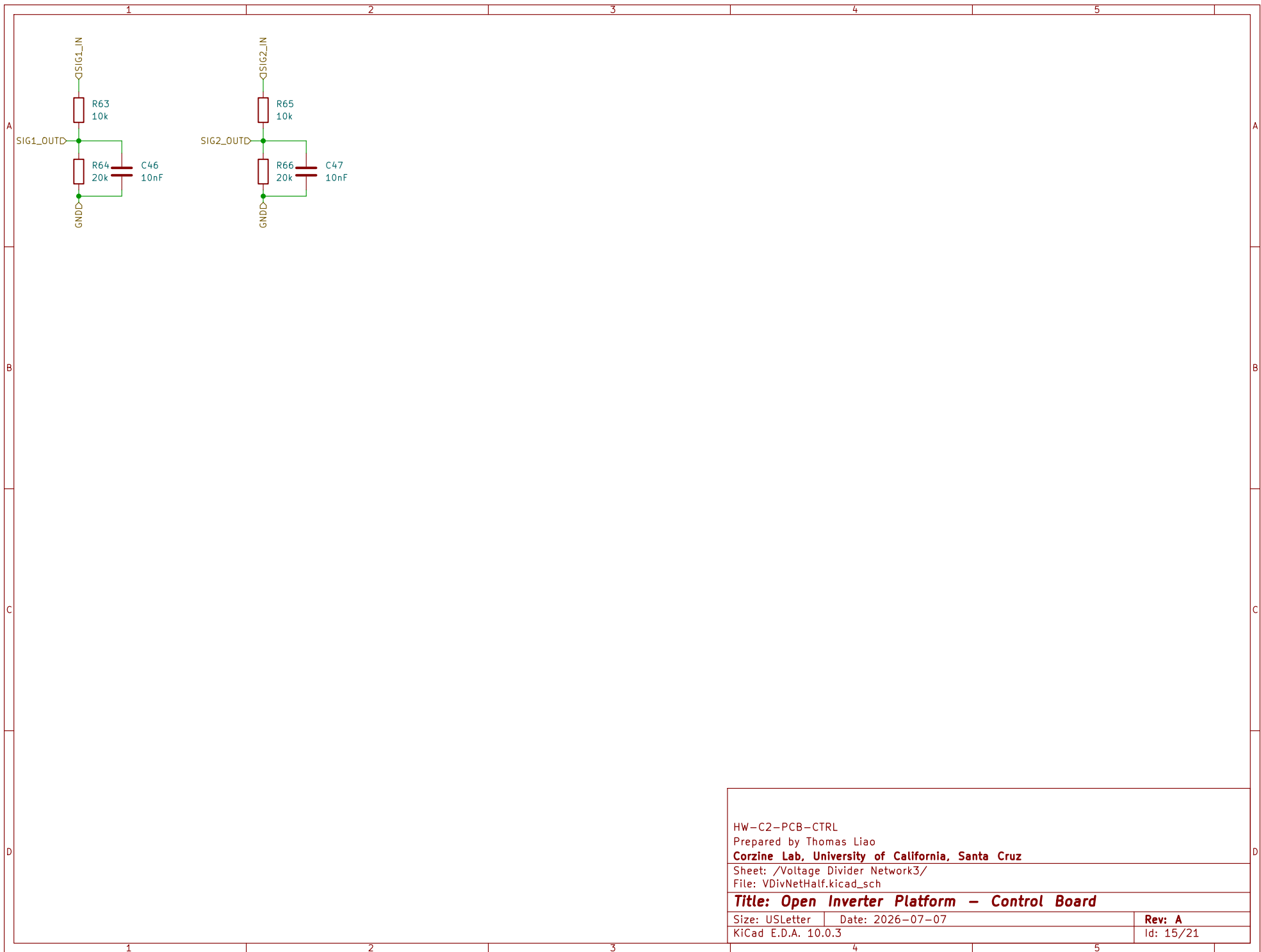
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Date: 2026-07-07

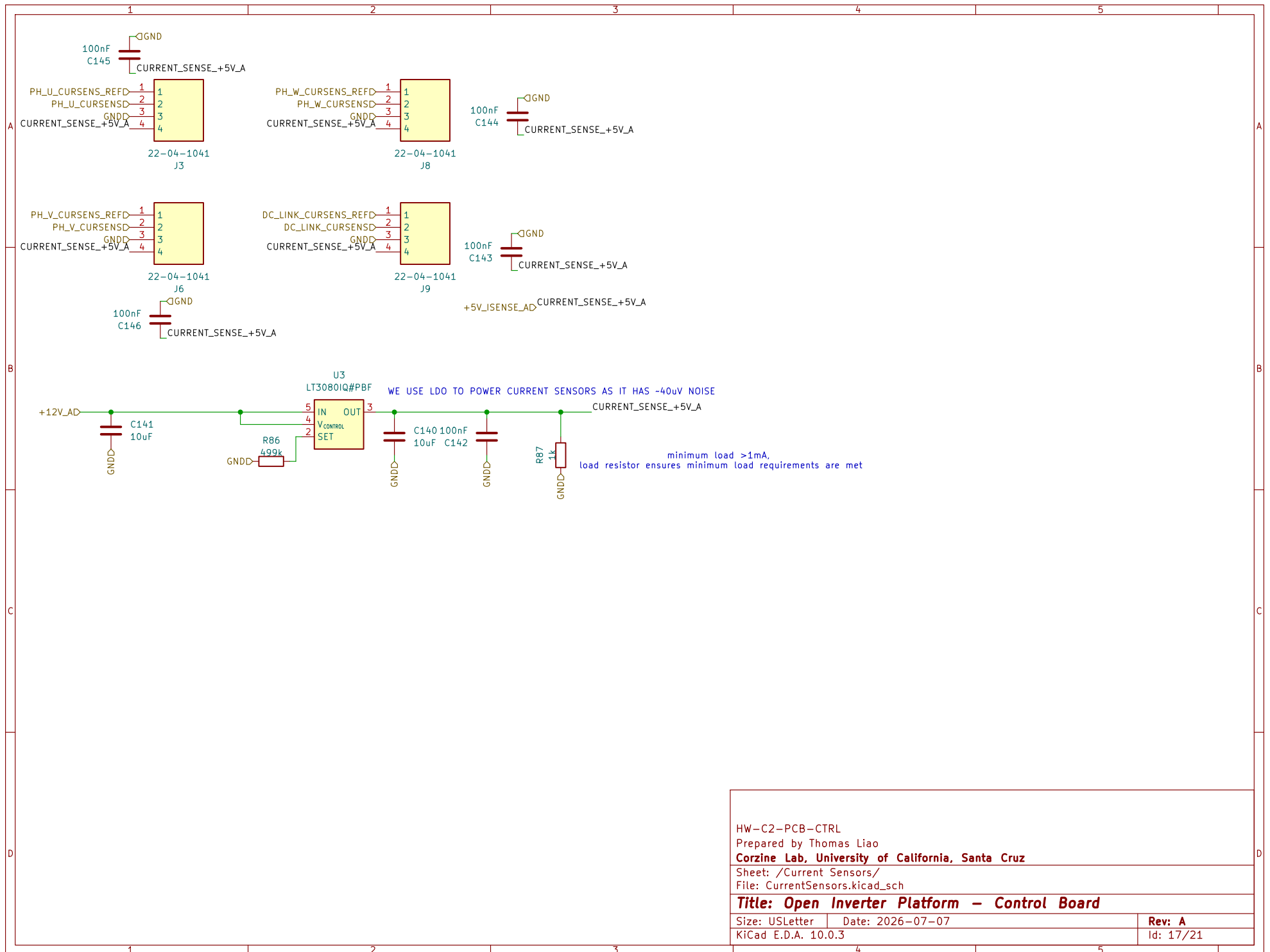
Rev: A

KiCad E.D.A. 10.0.3

Id: 14/21



HW-C2-PCB-CTRL		
Prepared by Thomas Liao		
Corzine Lab, University of California, Santa Cruz		
Sheet: /Voltage Divider Network3/		
File: VDivNetHalf.kicad_sch		
Title: Open Inverter Platform – Control Board		
Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3	Id: 15/21	



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Current Sensors/

File: CurrentSensors.kicad_sch

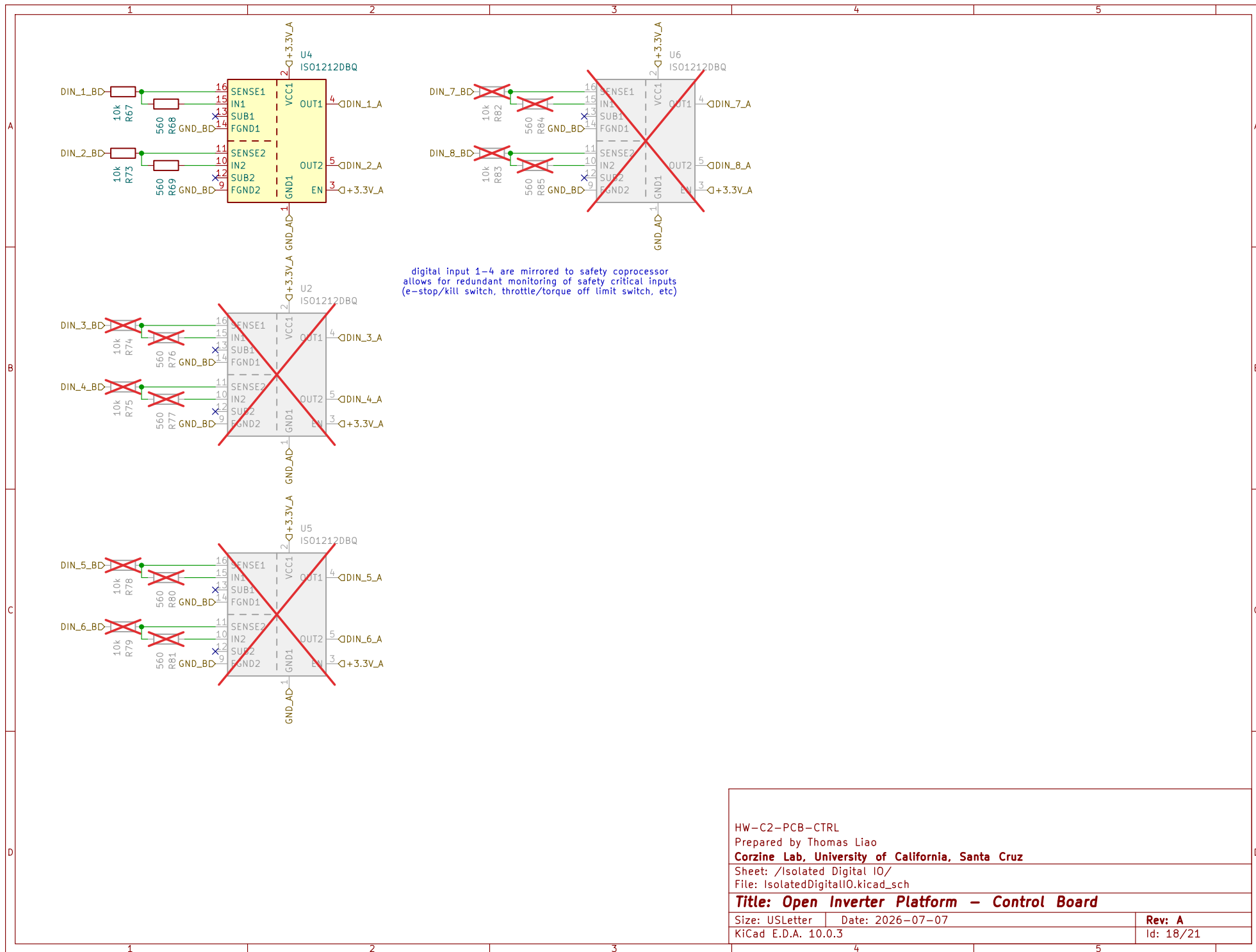
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Size: USLetter Date: 2026-07-07

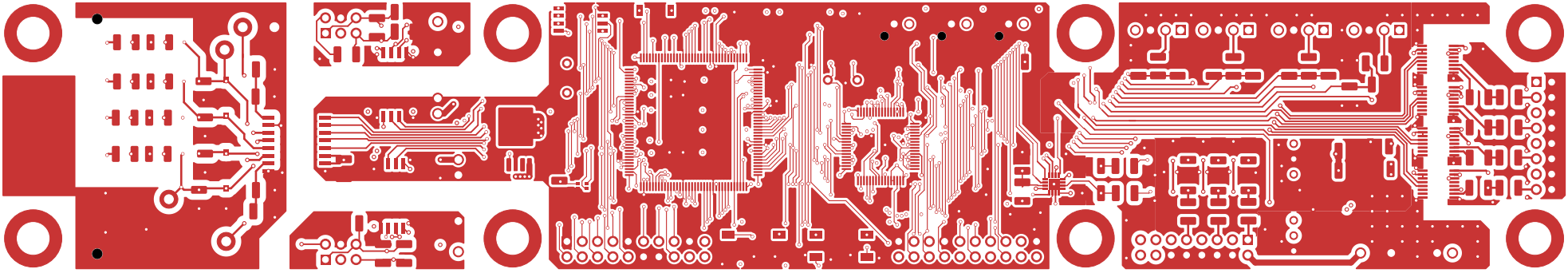
KiCad E.D.A. 10.0.3

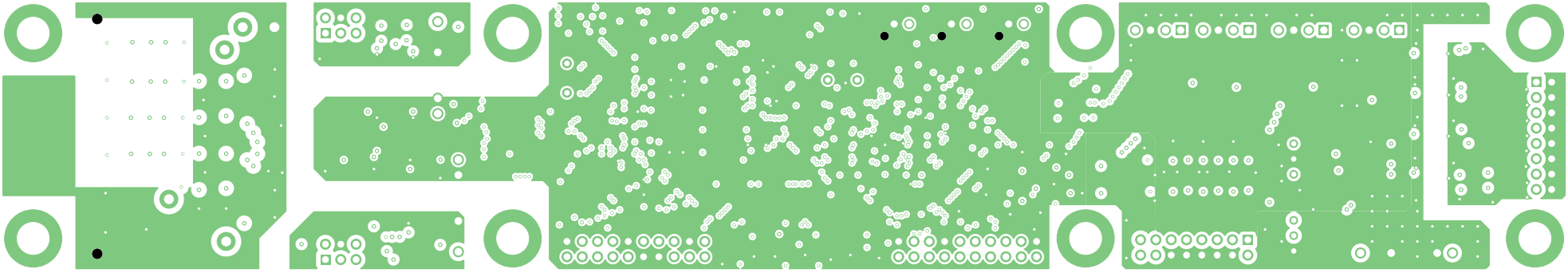
Rev: A

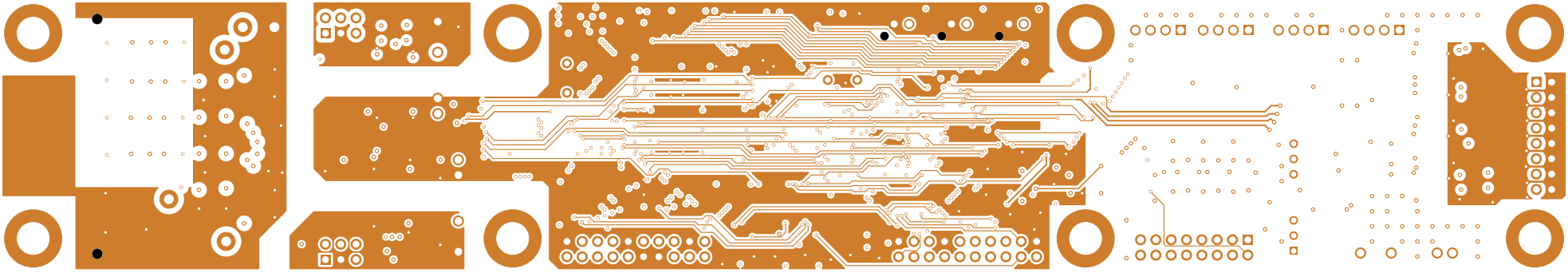
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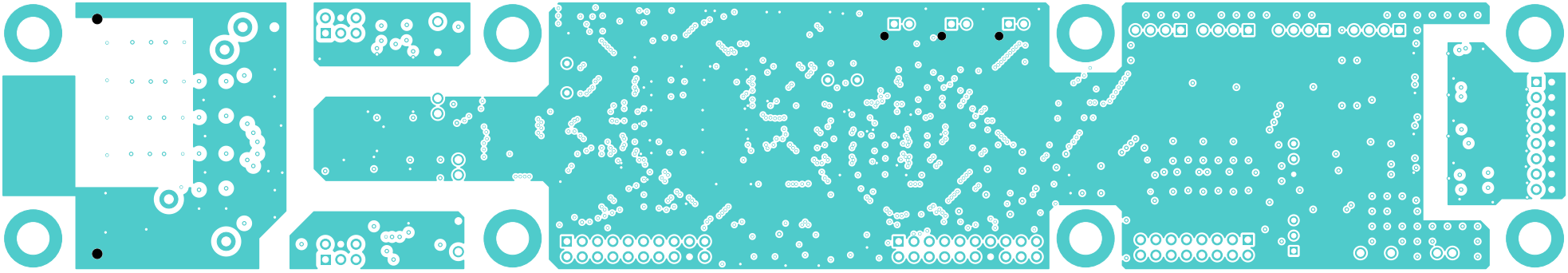


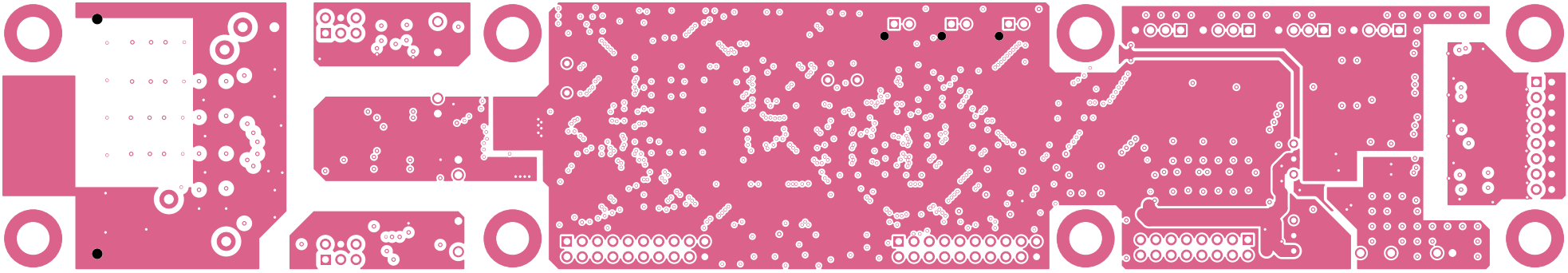
HW-C2-PCB-CTRL		
Prepared by Thomas Liao		
Corzine Lab, University of California, Santa Cruz		
Sheet: /Isolated Digital IO/		
File: IsolatedDigitalIO.kicad_sch		
Title: Open Inverter Platform - Control Board		
Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3		Id: 18/21

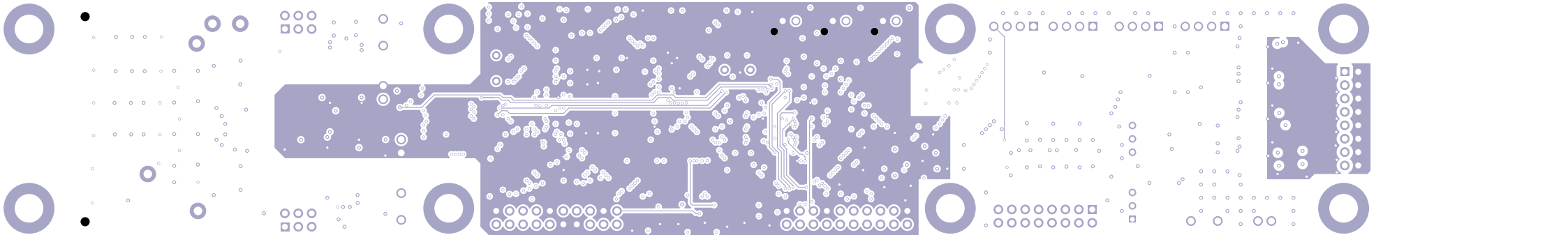


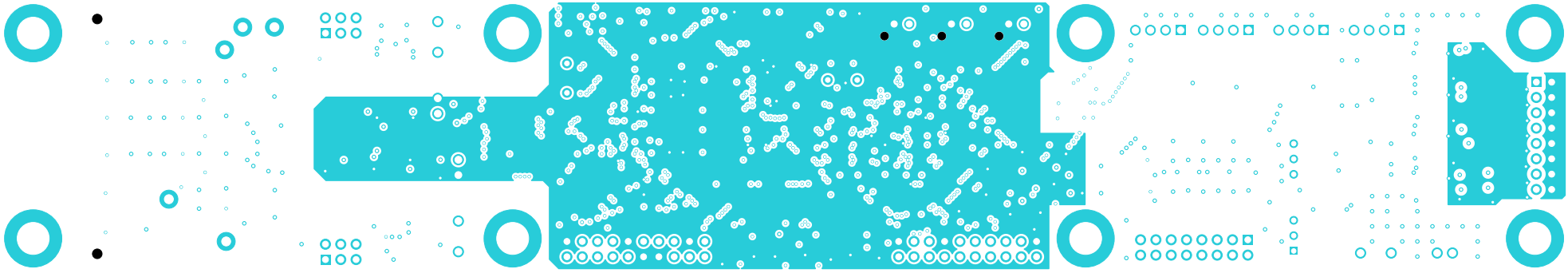


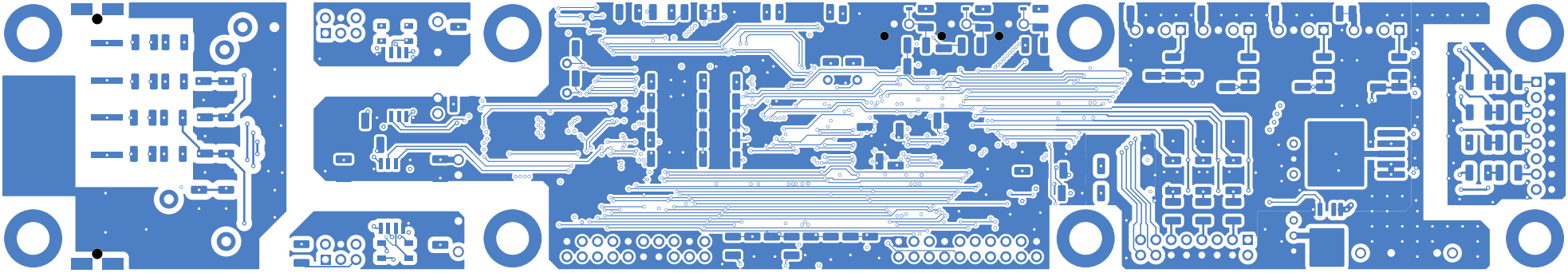


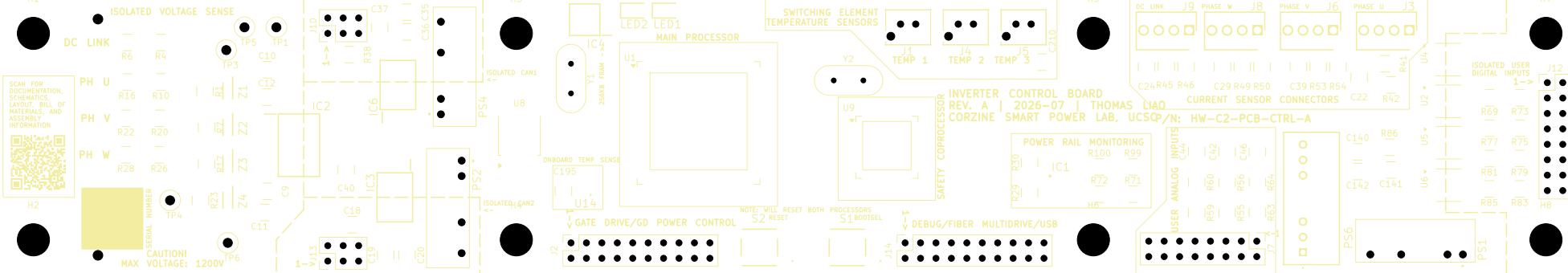






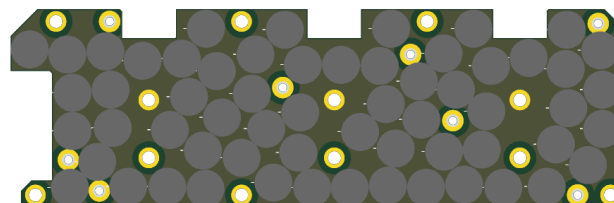
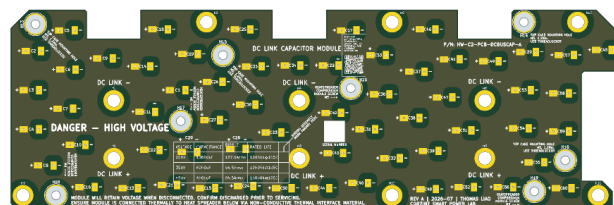


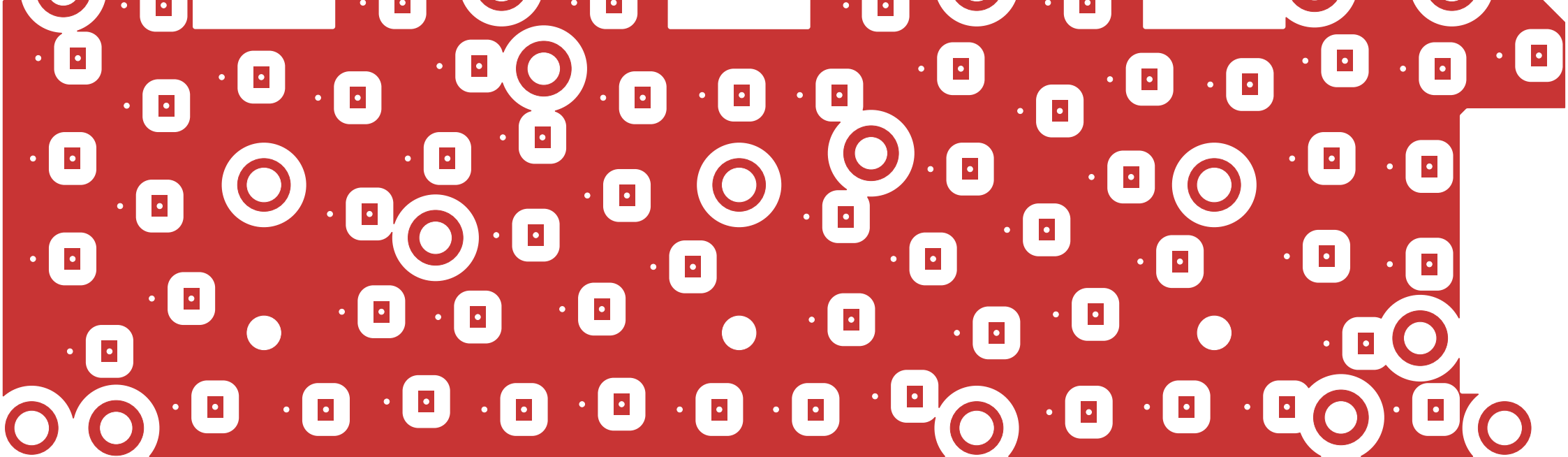


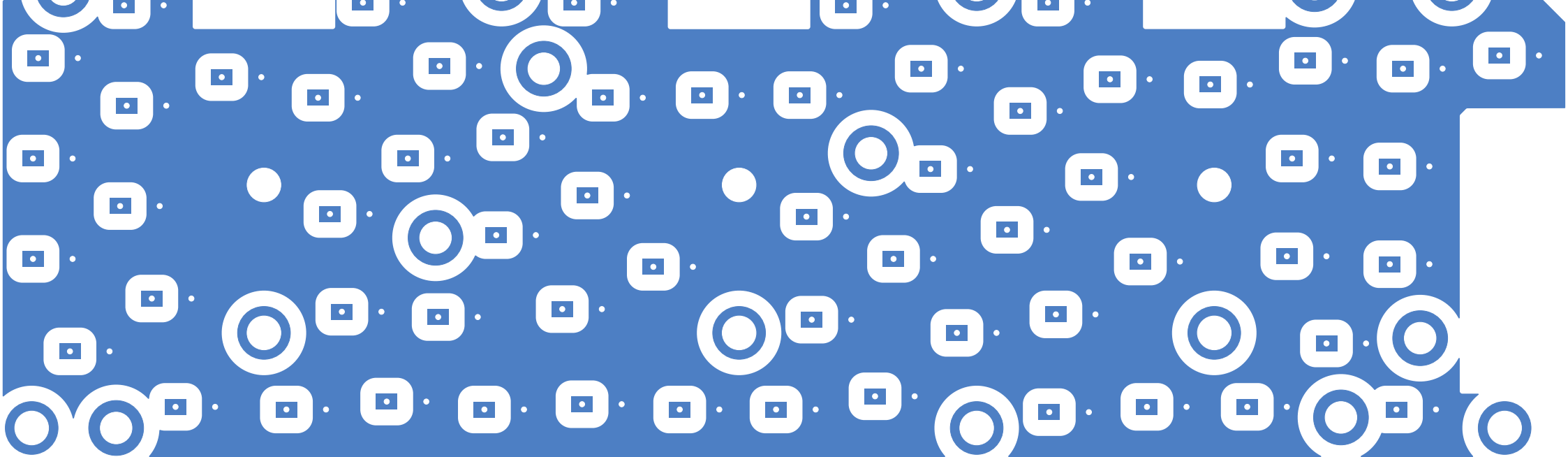


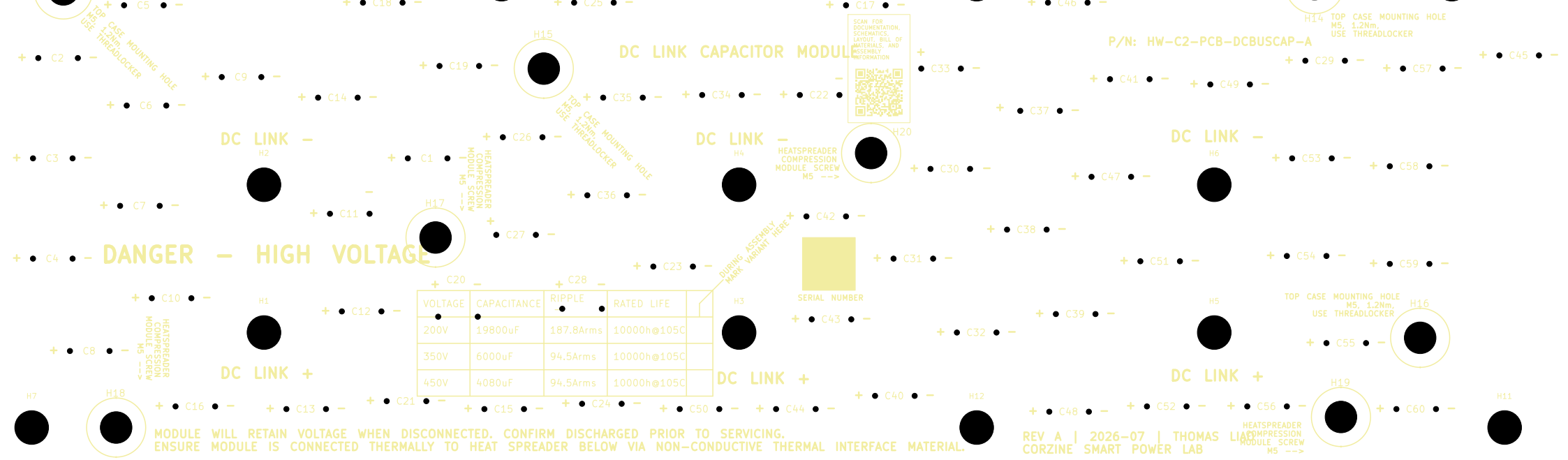
DCBUSCAPACITORBOARD

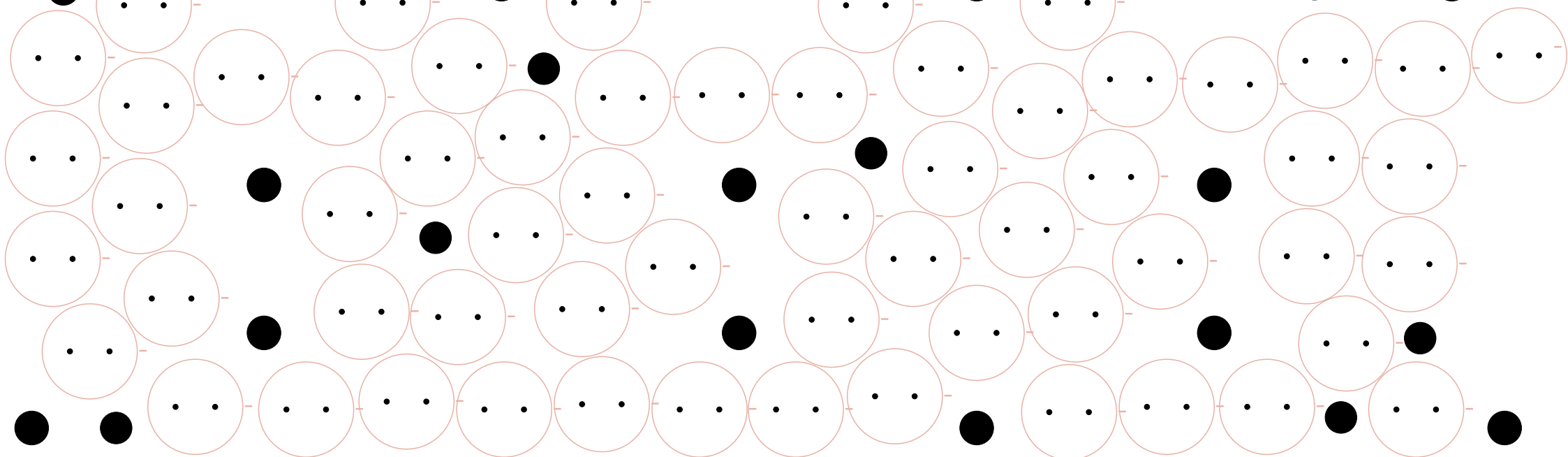
HW-C2-PCB-DCBUSCAP-A · printed circuit board — schematic and PCB layers





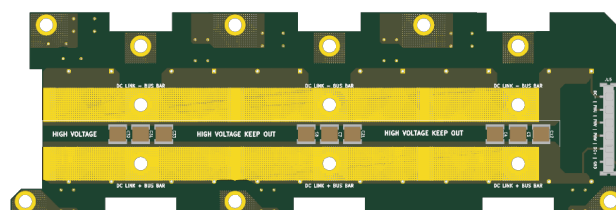
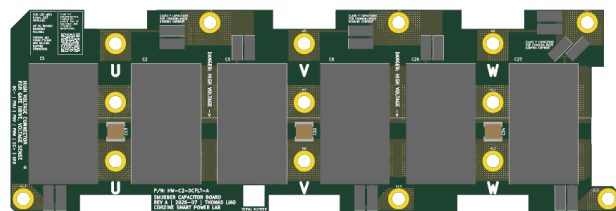


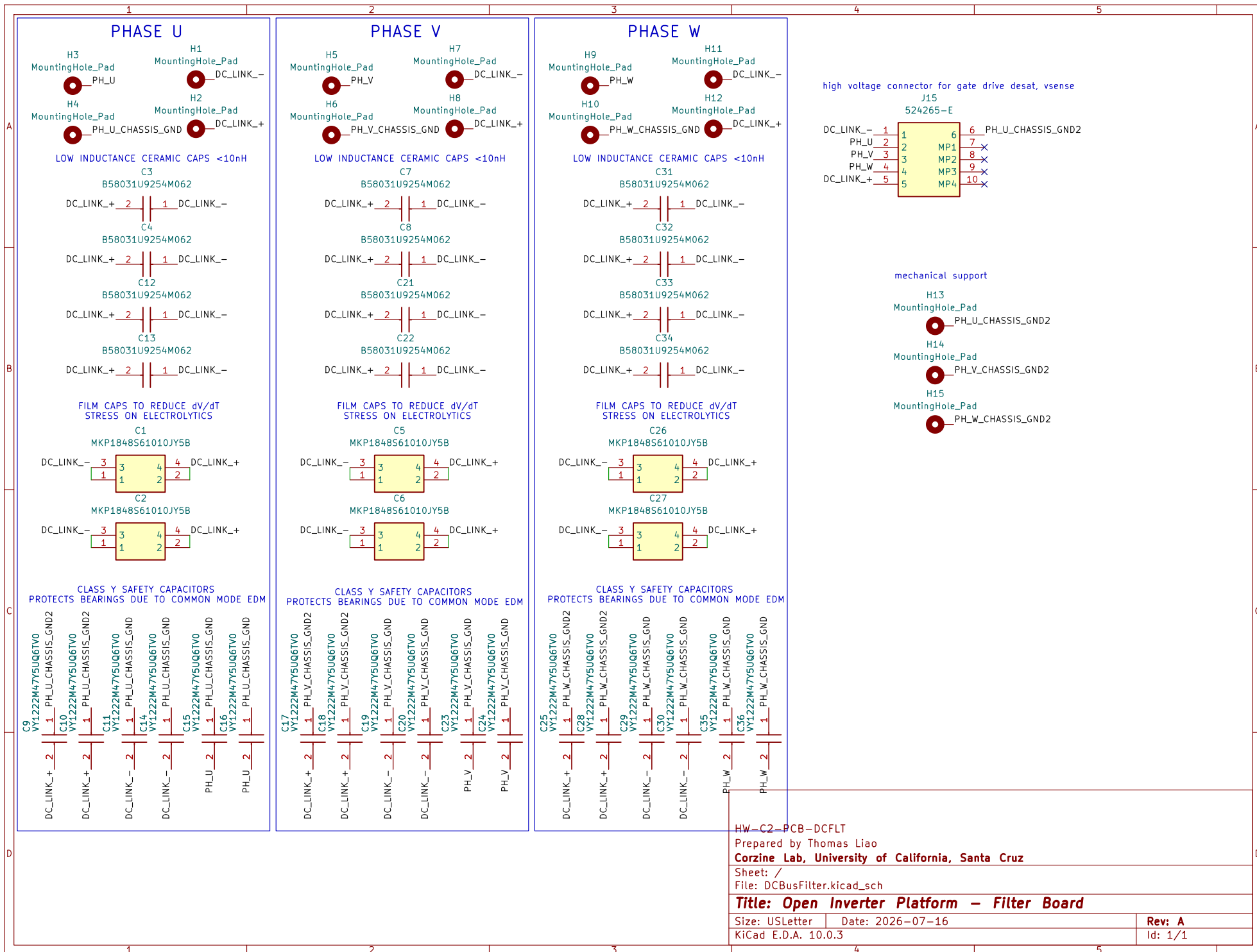


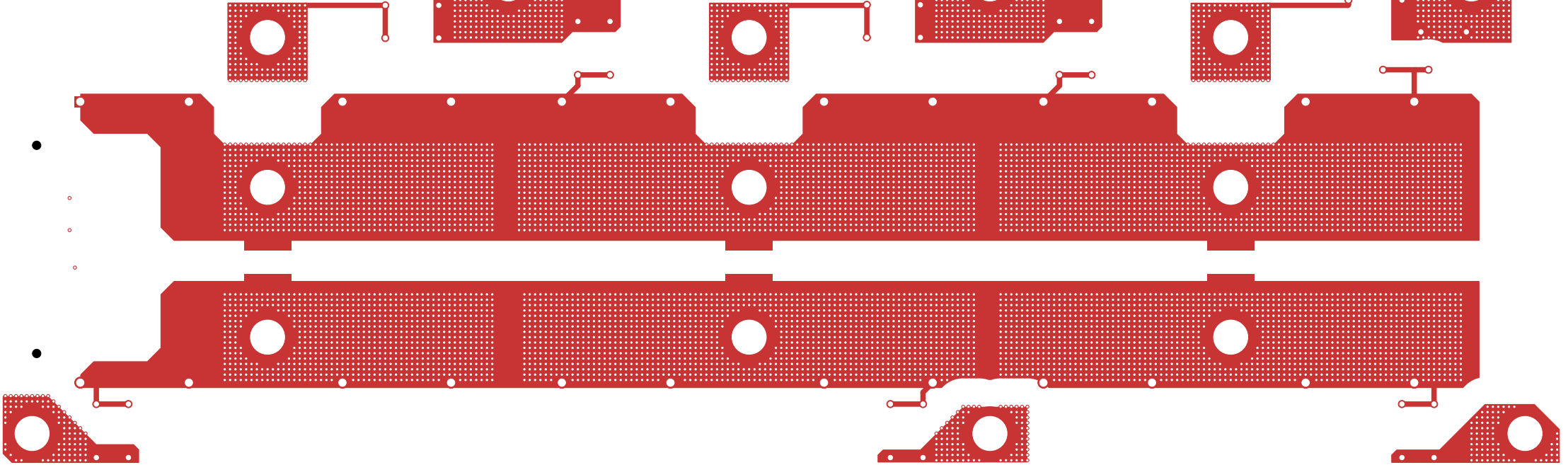


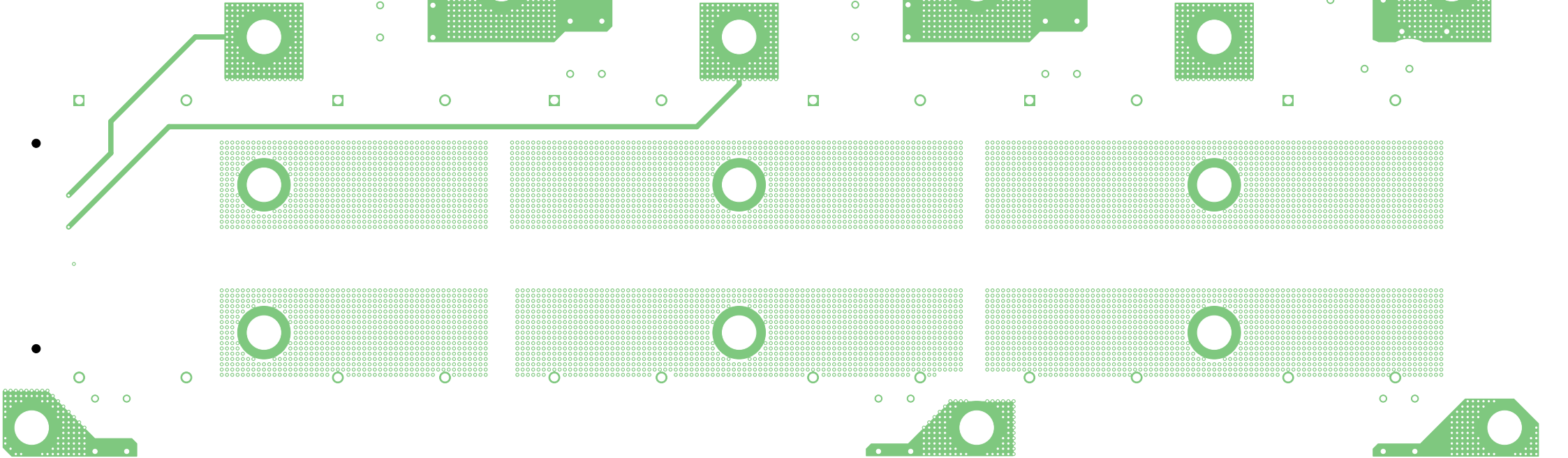
DCBUSFILTER

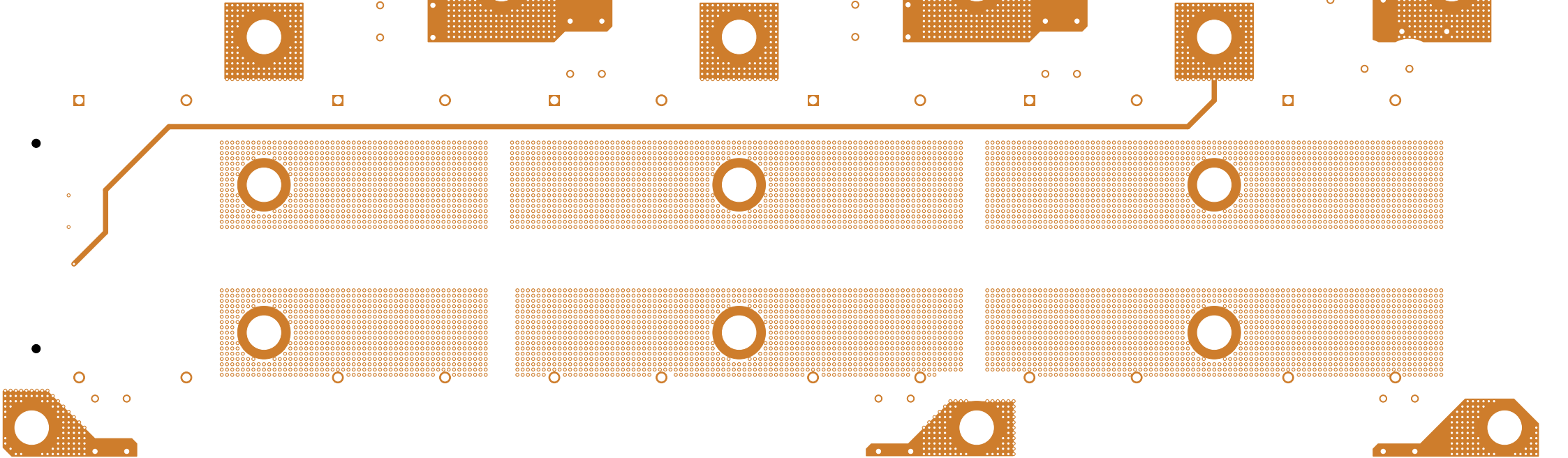
HW-C2-PCB-DCFLT-A · printed circuit board — schematic and PCB layers

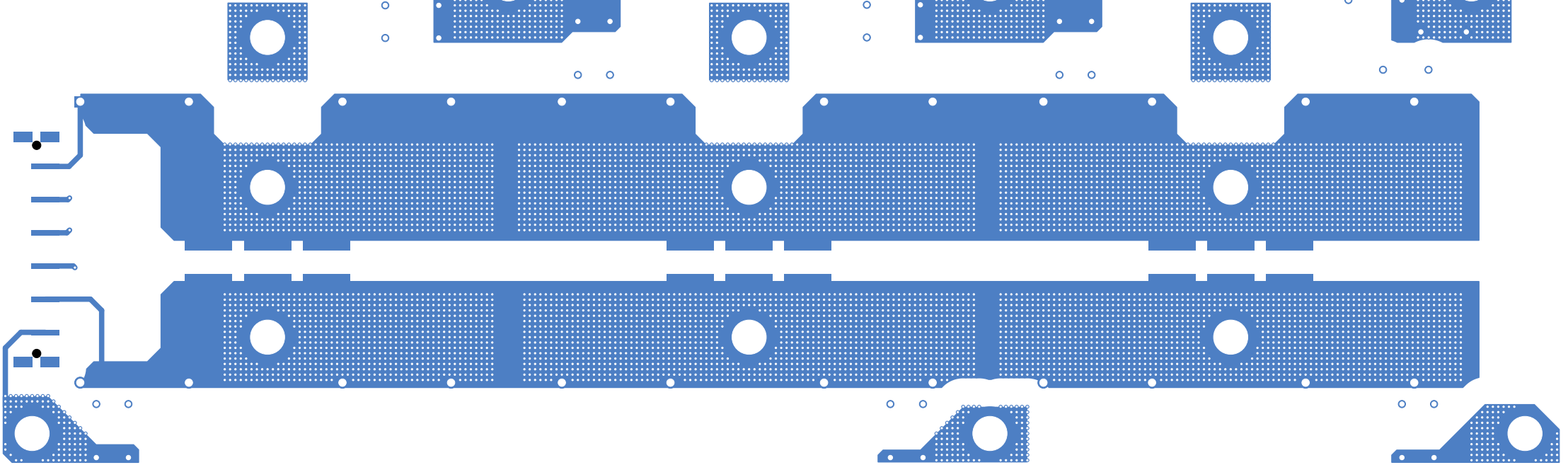


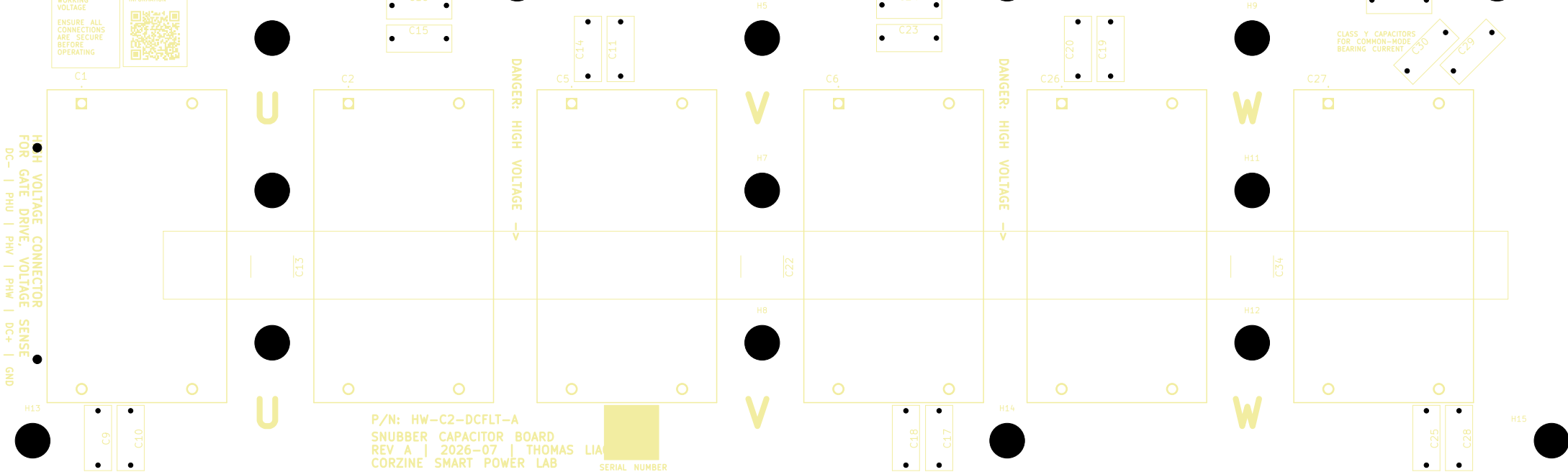


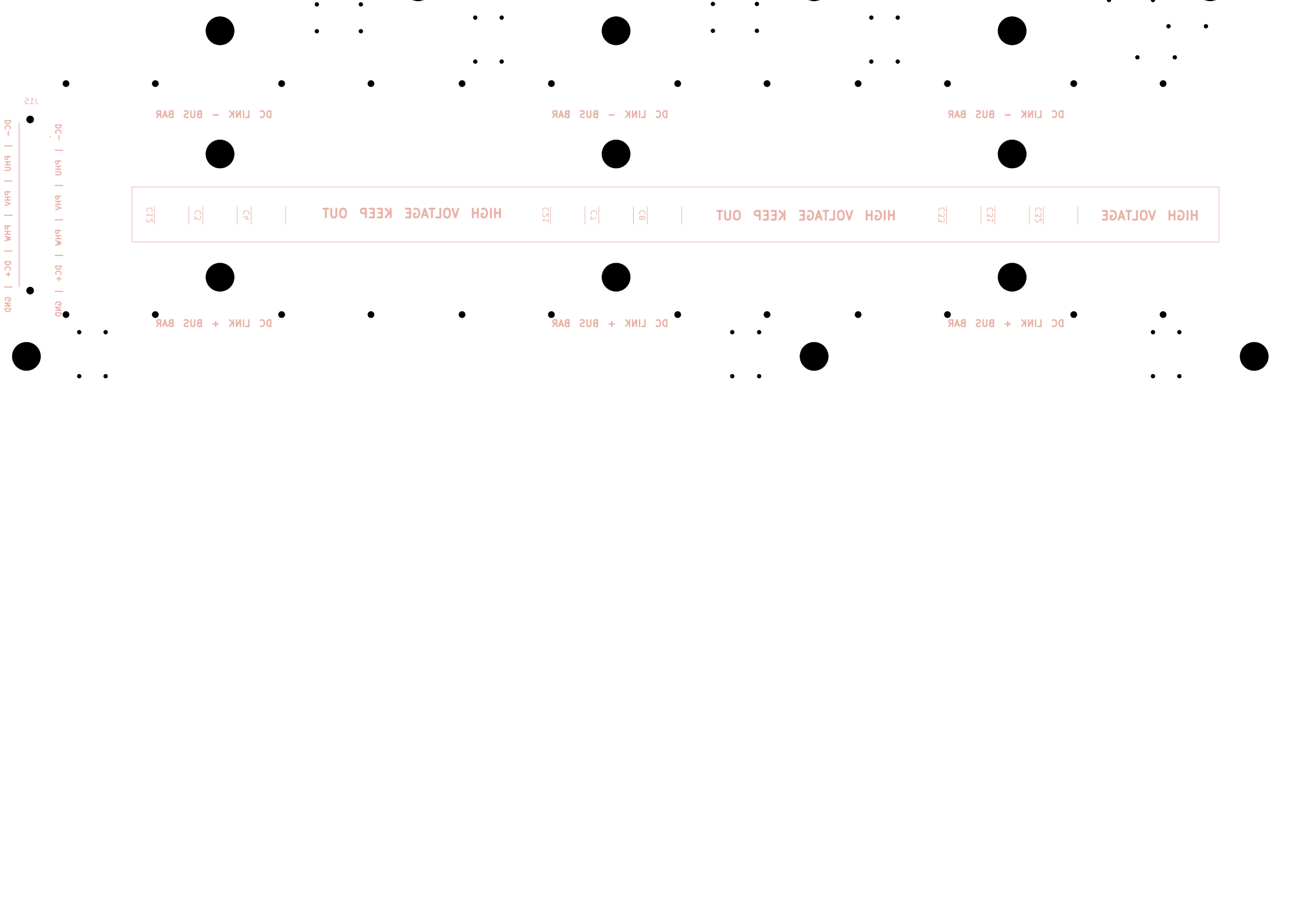






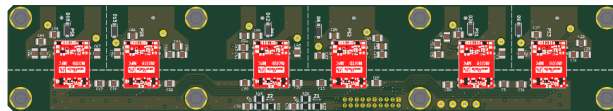
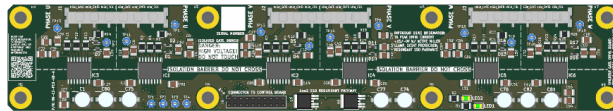


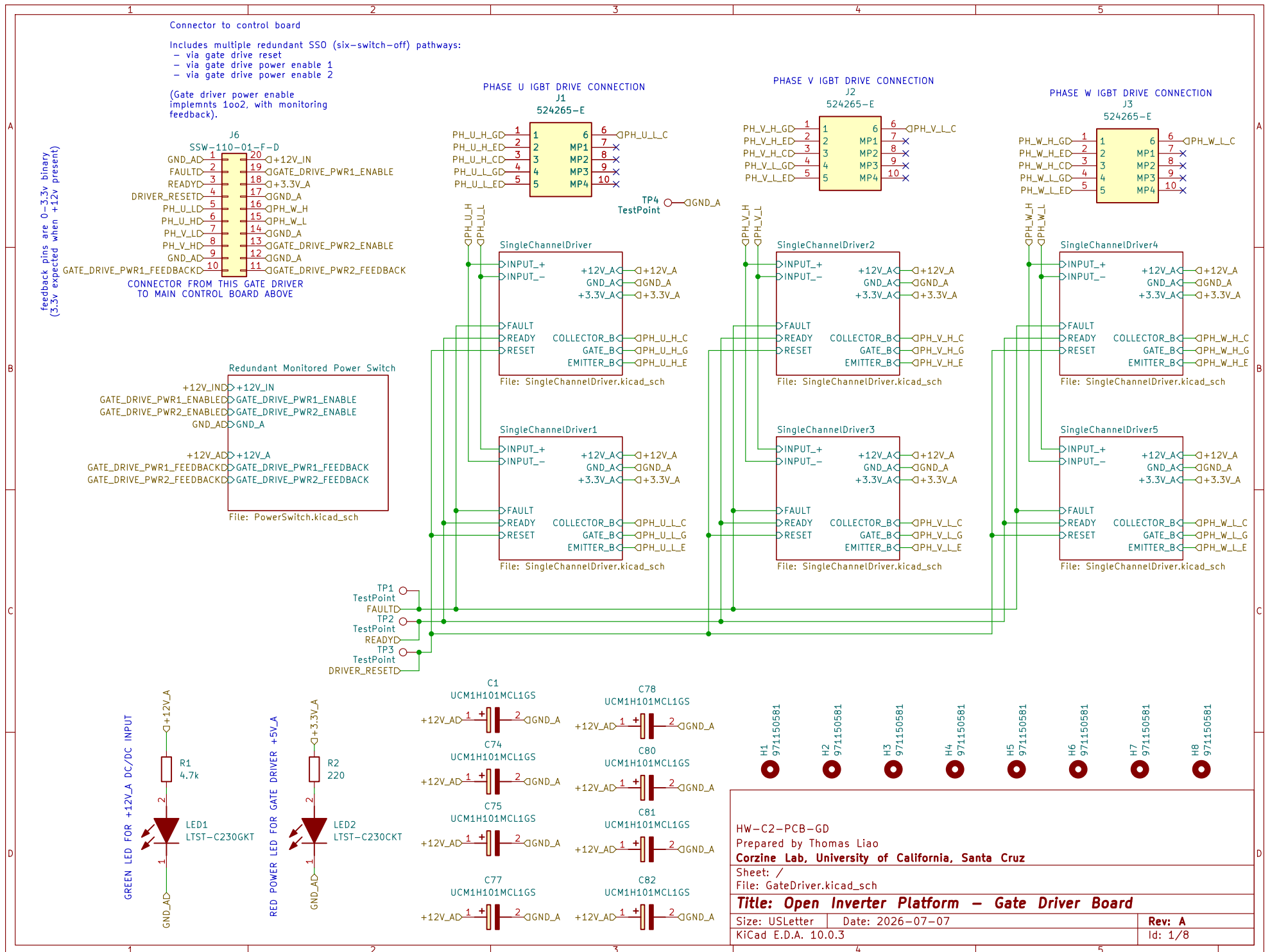


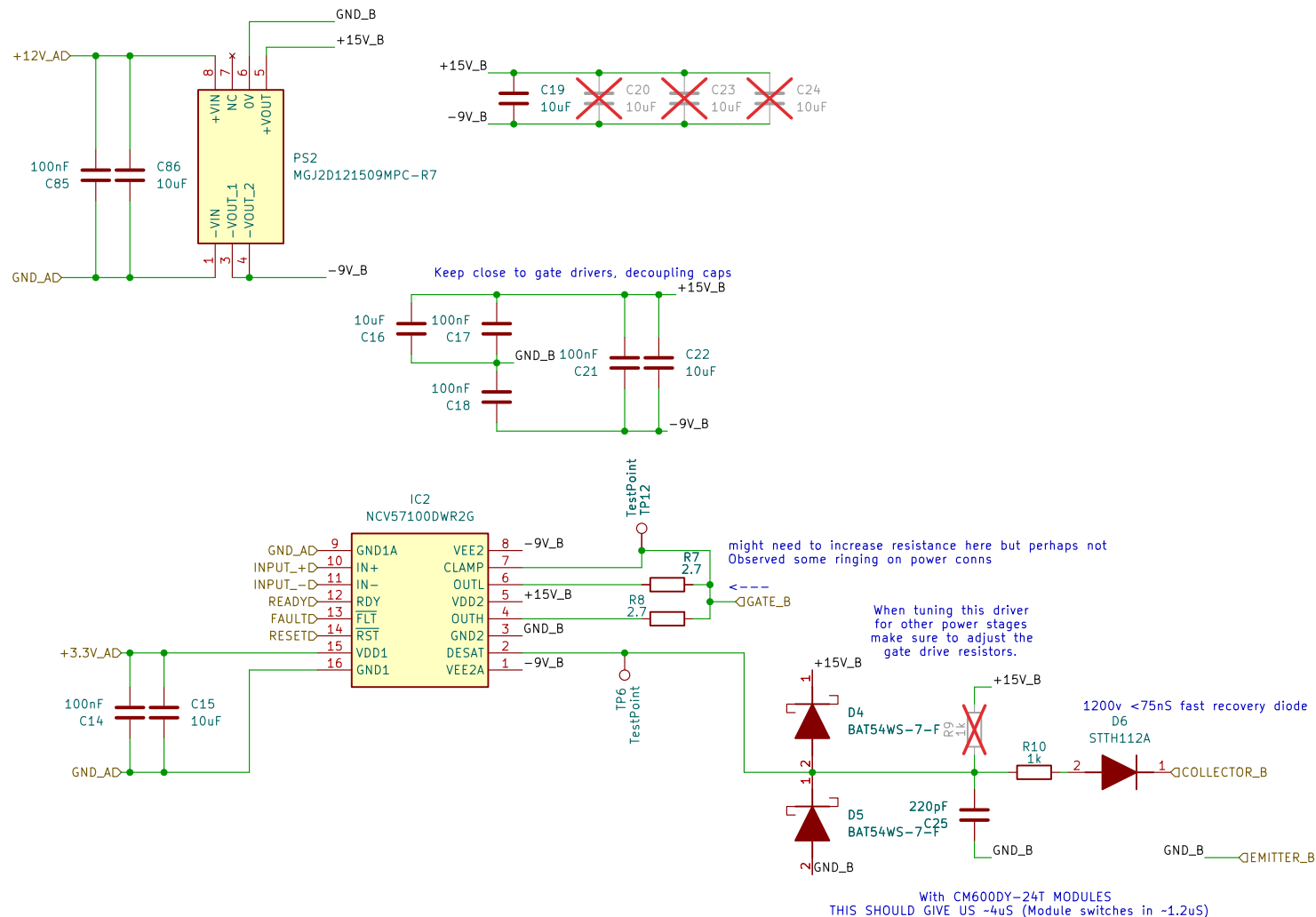


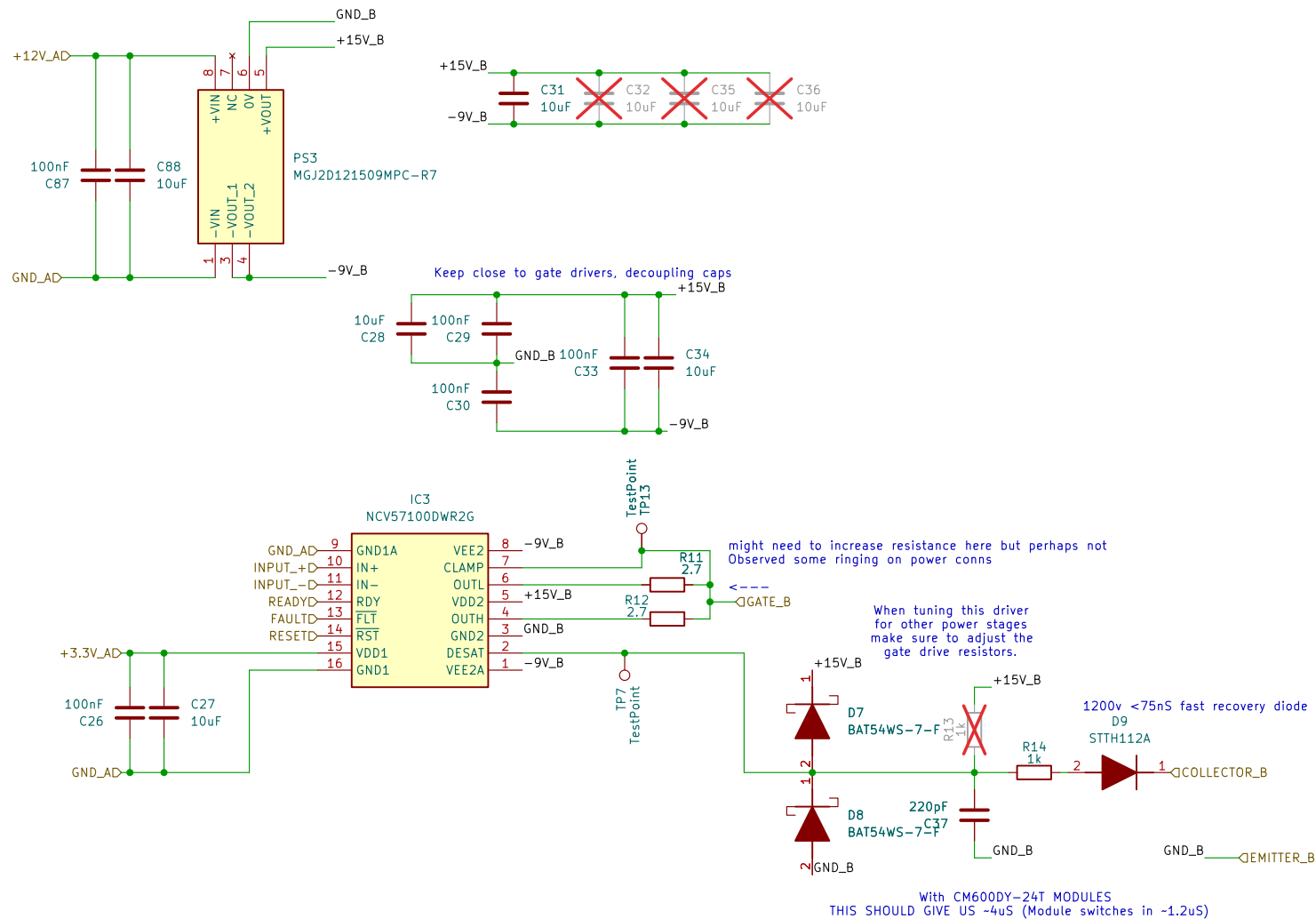
GATEDRIVER

HW-C2-PCB-GD-A · printed circuit board — schematic and PCB layers









HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver/

File: SingleChannelDriver.kicad_sch

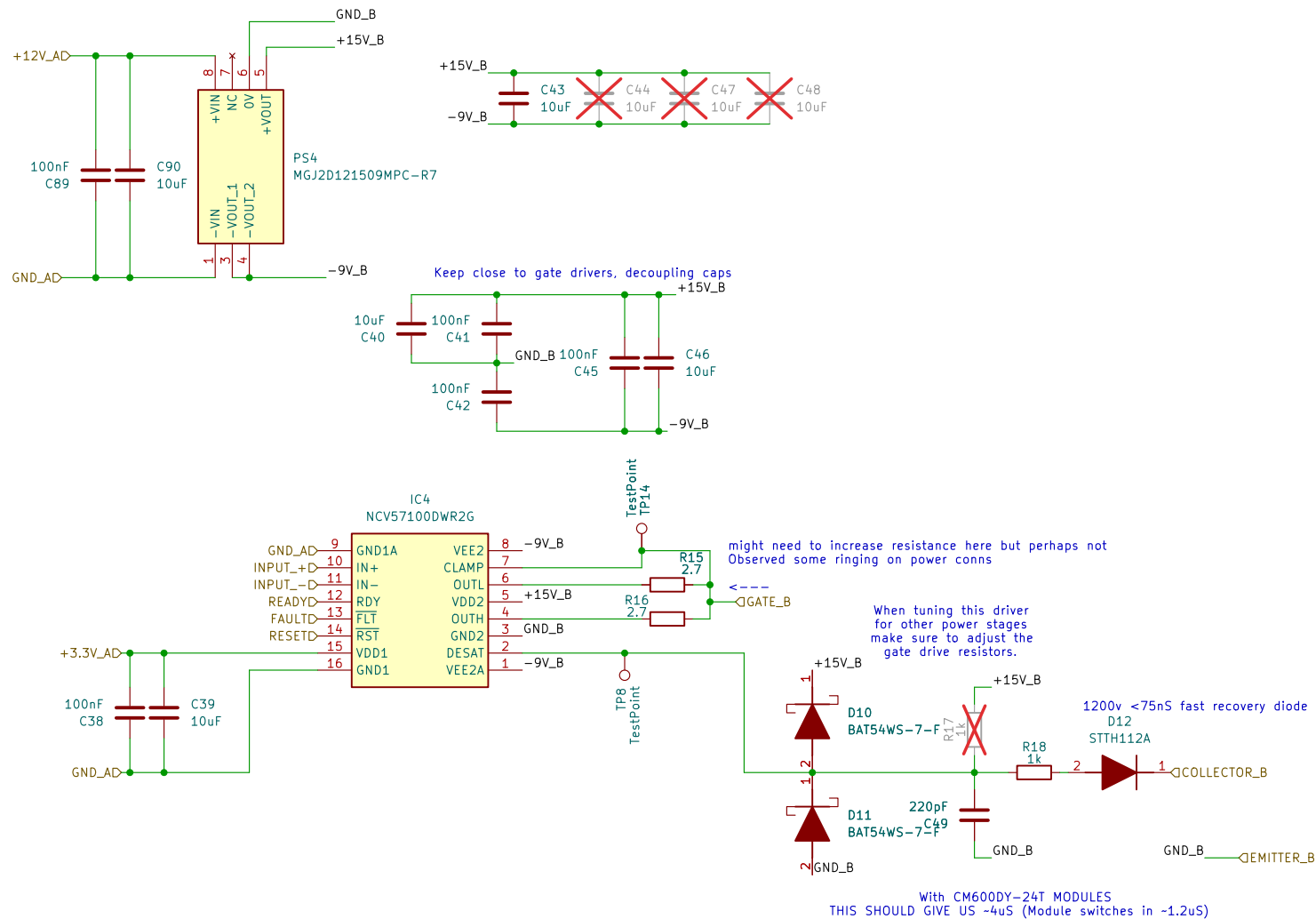
Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 4/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver3/

File: SingleChannelDriver.kicad_sch

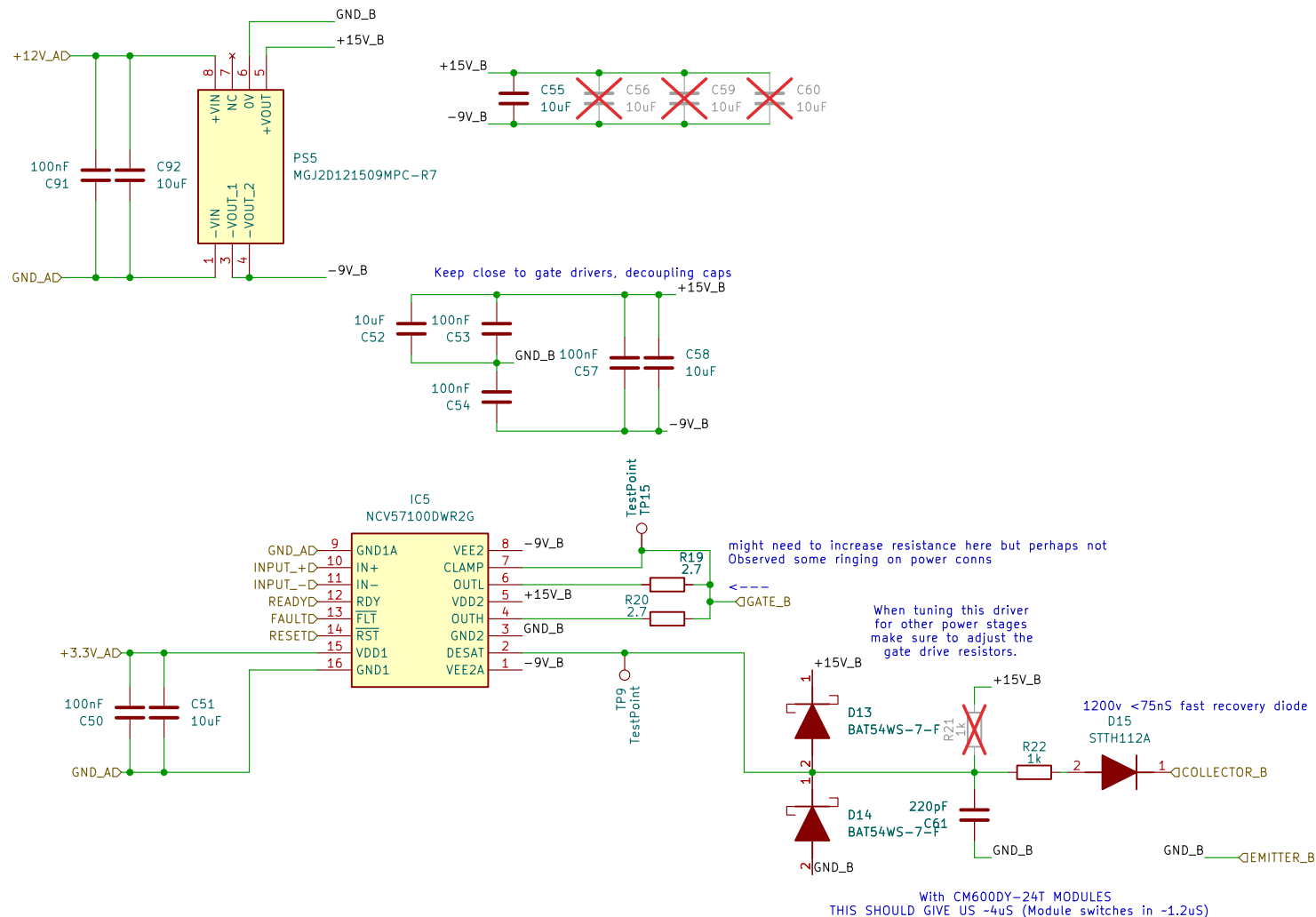
Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 5/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver4/

File: SingleChannelDriver.kicad_sch

Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

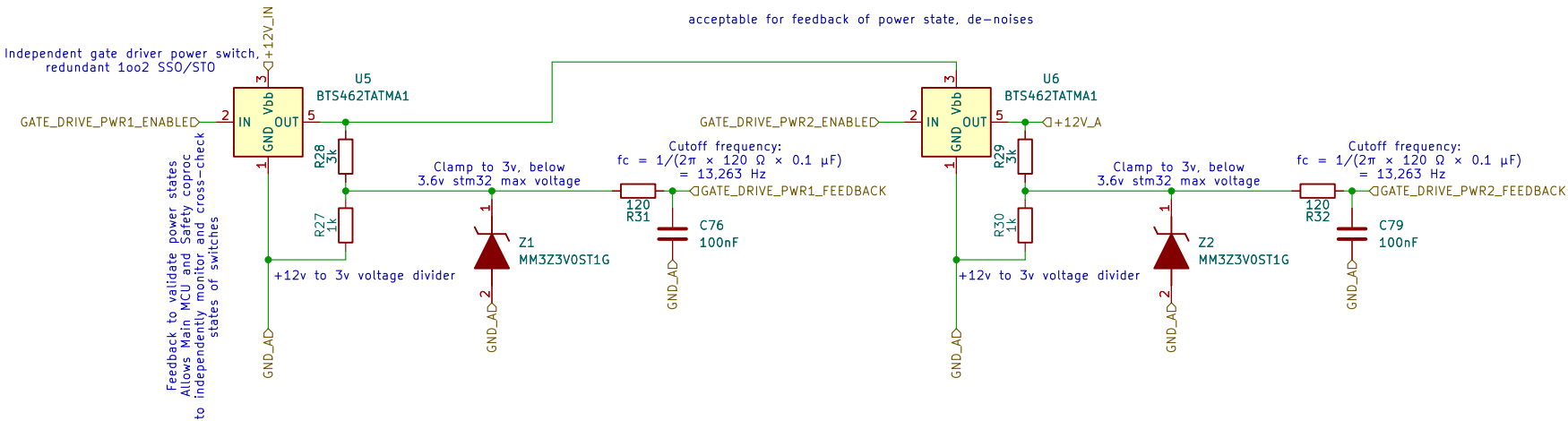
Id: 6/8

This circuit implements a redundant pathway for both mcus to implement Six Switch Open / Safe Torque Off in addition to regular pwm break and gate driver reset.

The following switches are independent systems with feedback monitoring. Each MCU cross monitors returned inputs, and can validate if the switch failed to respond as commanded – thus removing latent failures from this safety subsystem.

cutoff period (1/freq) ->
75.4 uS

acceptable for feedback of power state, de-noises



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Redundant Monitored Power Switch/

File: PowerSwitch.kicad_sch

Title: Open Inverter Platform – Gate Driver Board

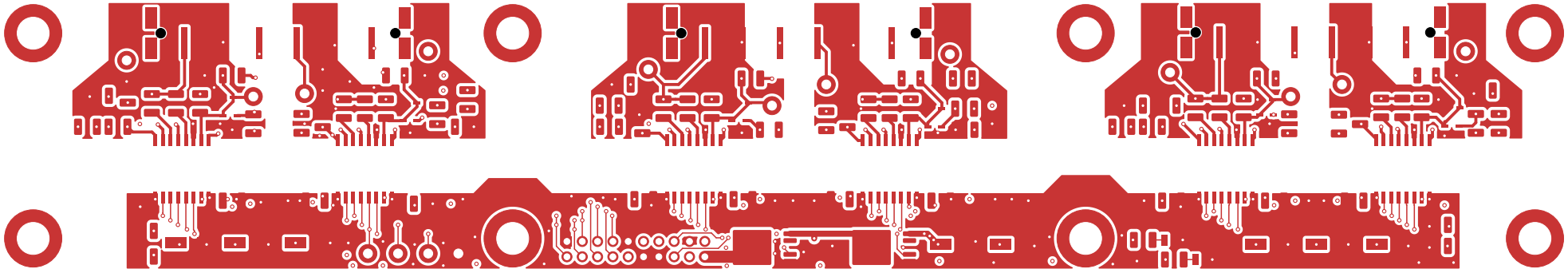
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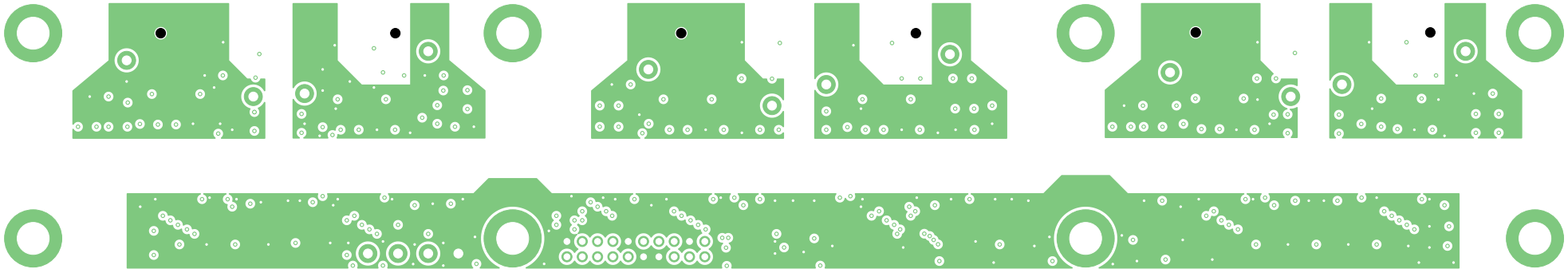
Date: 2026-07-07

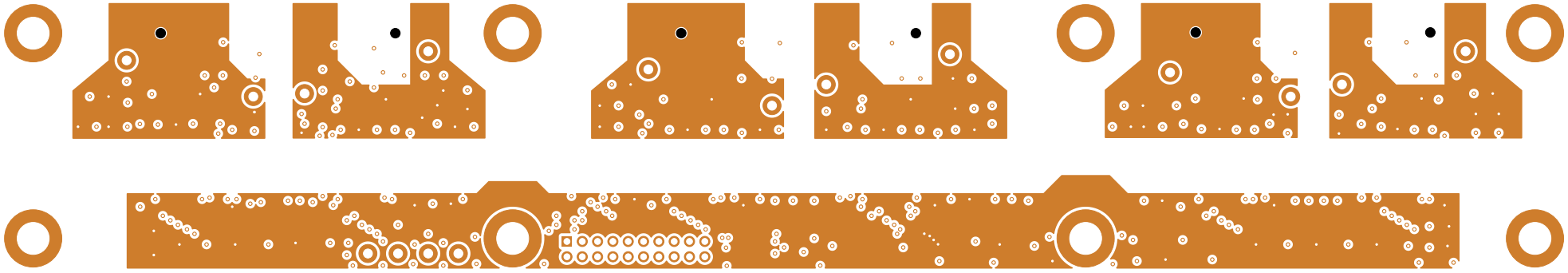
Rev: A

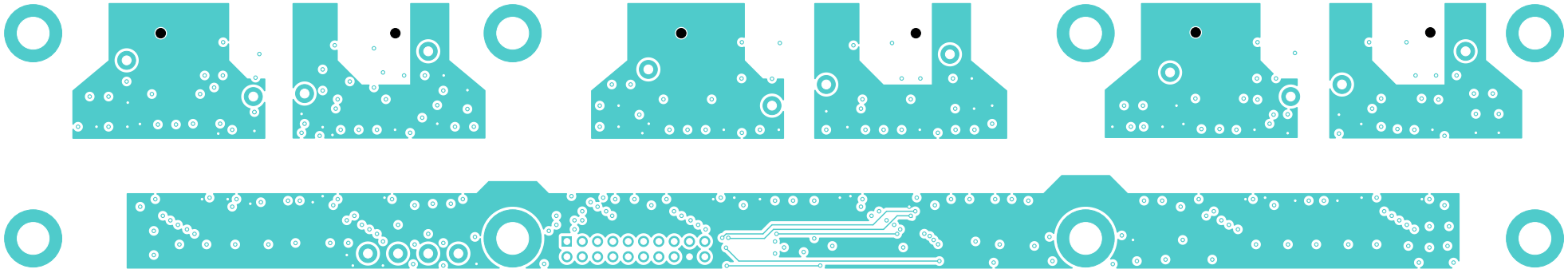
KiCad E.D.A. 10.0.3

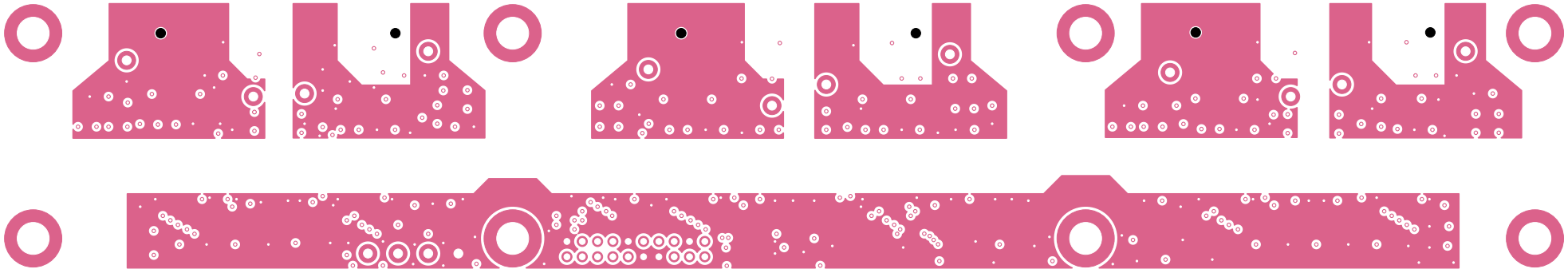
Id: 8/8

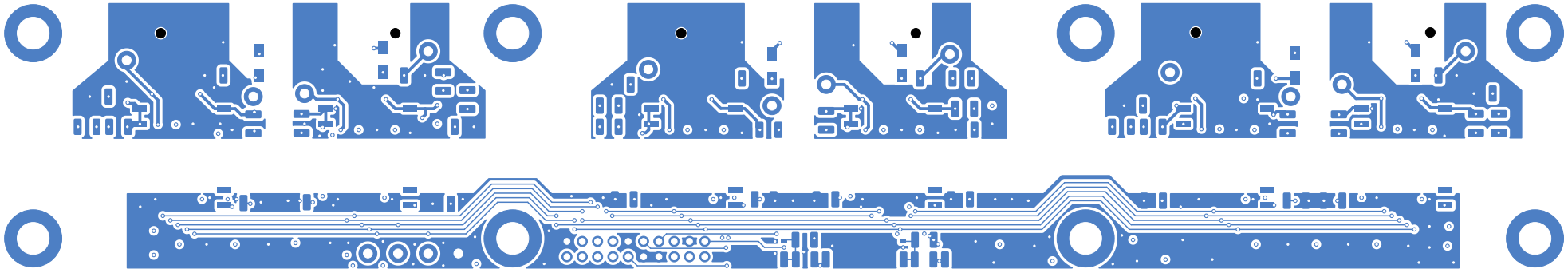


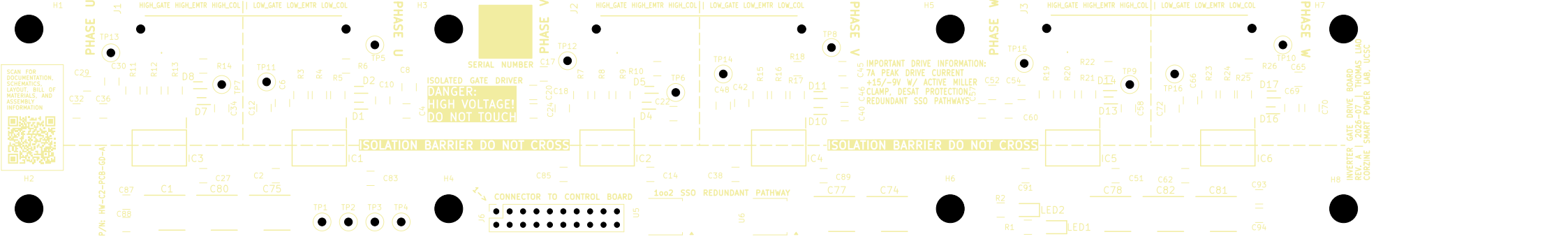


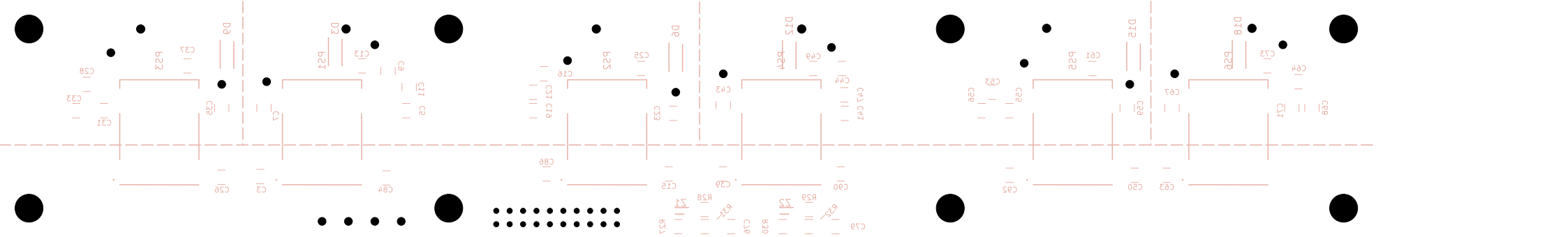






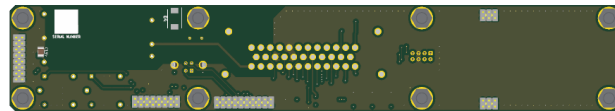
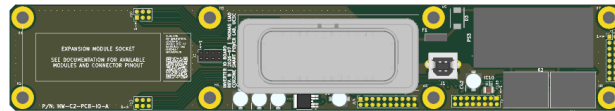


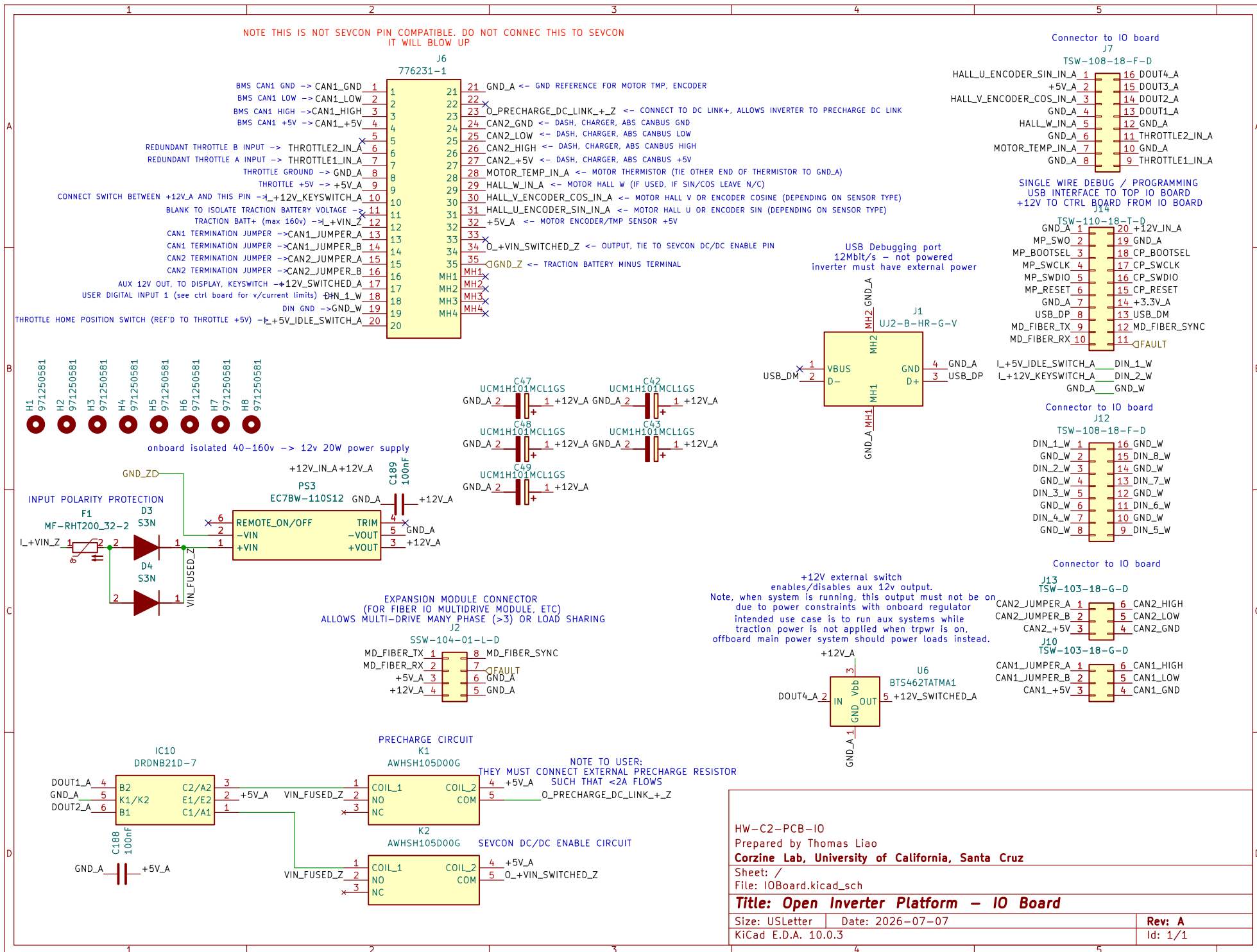


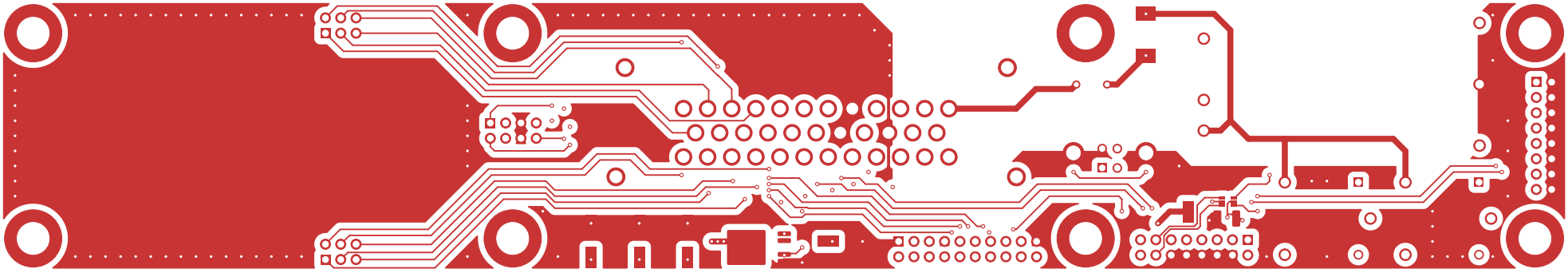


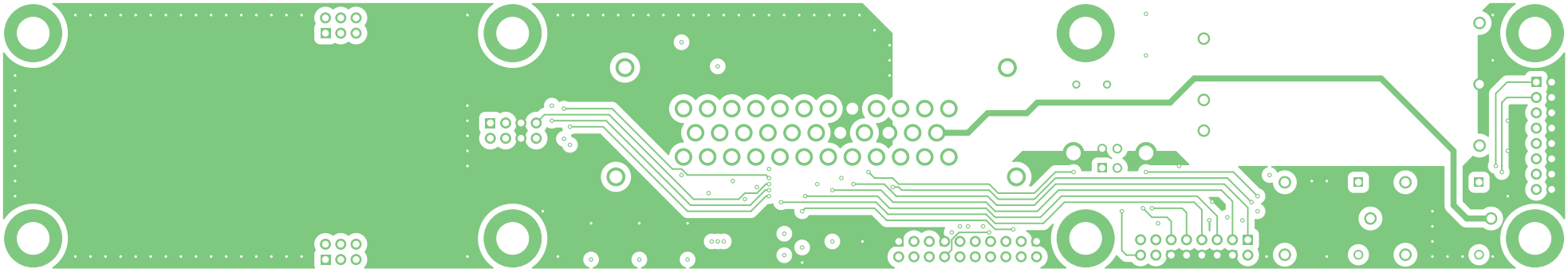
IOBOARD

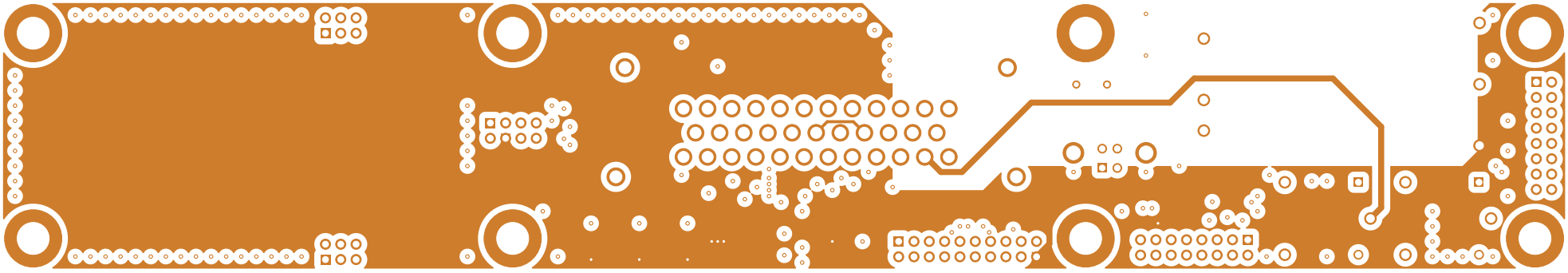
HW-C2-PCB-IO-A · printed circuit board — schematic and PCB layers

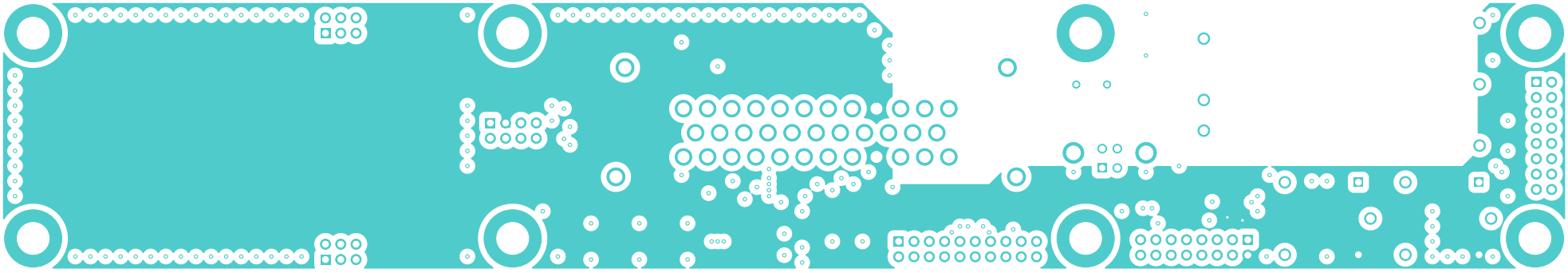


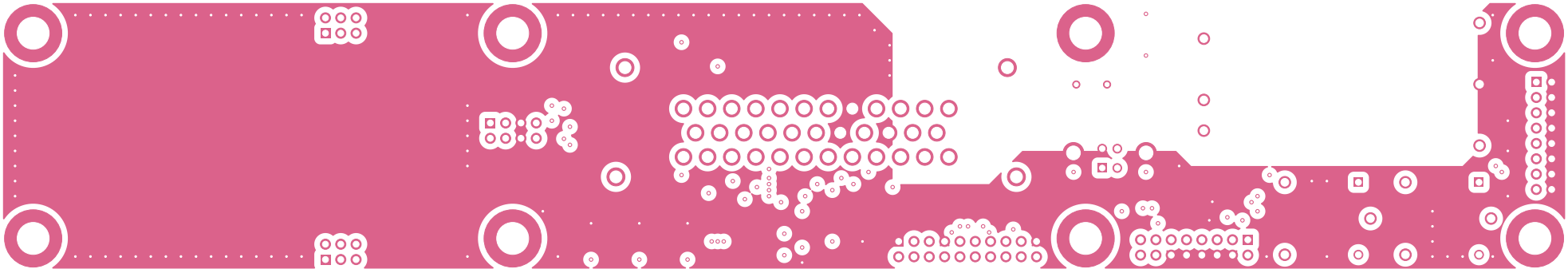


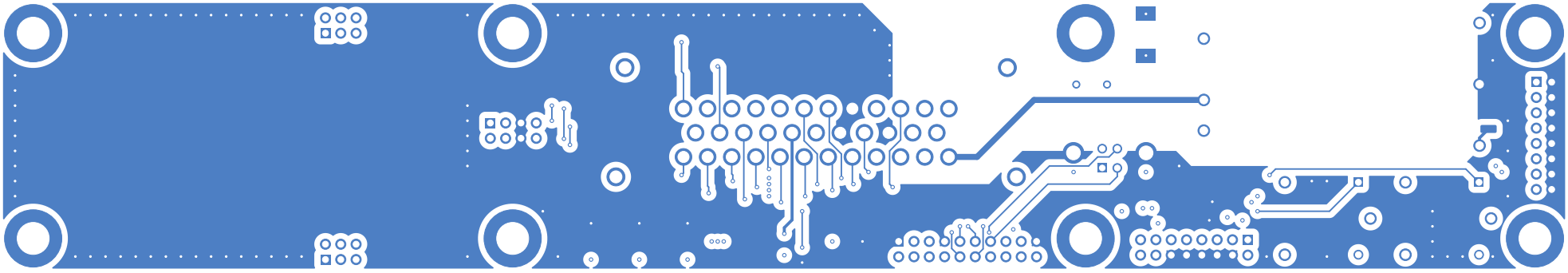


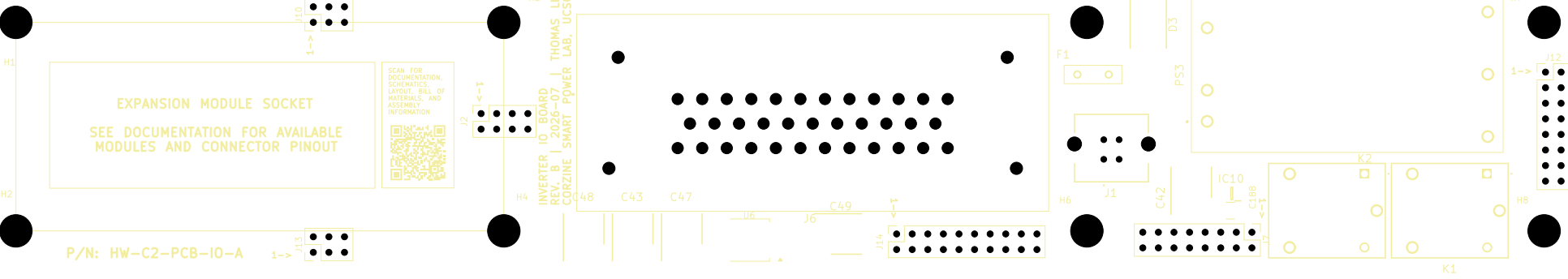


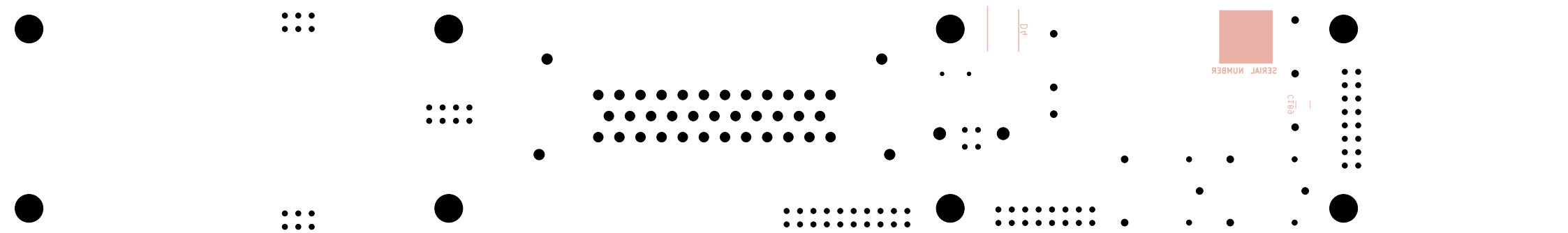












CAPACITORTEMPSENSEHARNESS

HW-C2-WH-TSBCM-A · wiring harness — schematic



Wiring Harness Note

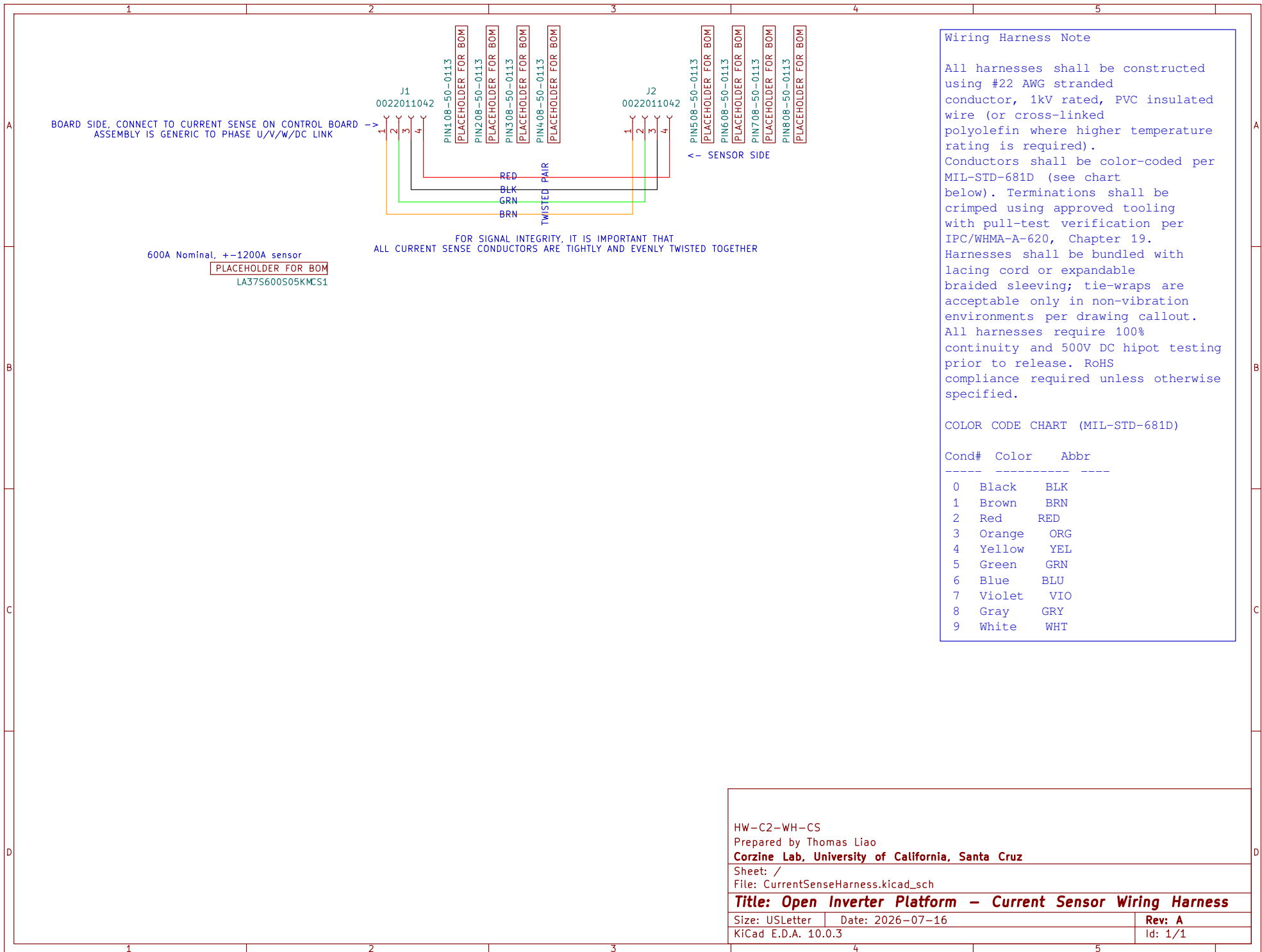
All harnesses shall be constructed using #22 AWG stranded conductor, 1kV rated, PVC insulated wire (or cross-linked polyolefin where higher temperature rating is required). Conductors shall be color-coded per MIL-STD-681D (see chart below). Terminations shall be crimped using approved tooling with pull-test verification per IPC/WHMA-A-620, Chapter 19. Harnesses shall be bundled with lacing cord or expandable braided sleeving; tie-wraps are acceptable only in non-vibration environments per drawing callout. All harnesses require 100% continuity and 500V DC hipot testing prior to release. RoHS compliance required unless otherwise specified.

COLOR CODE CHART (MIL-STD-681D)

Cond#	Color	Abbr
0	Black	BLK
1	Brown	BRN
2	Red	RED
3	Orange	ORG
4	Yellow	YEL
5	Green	GRN
6	Blue	BLU
7	Violet	VIO
8	Gray	GRY
9	White	WHT

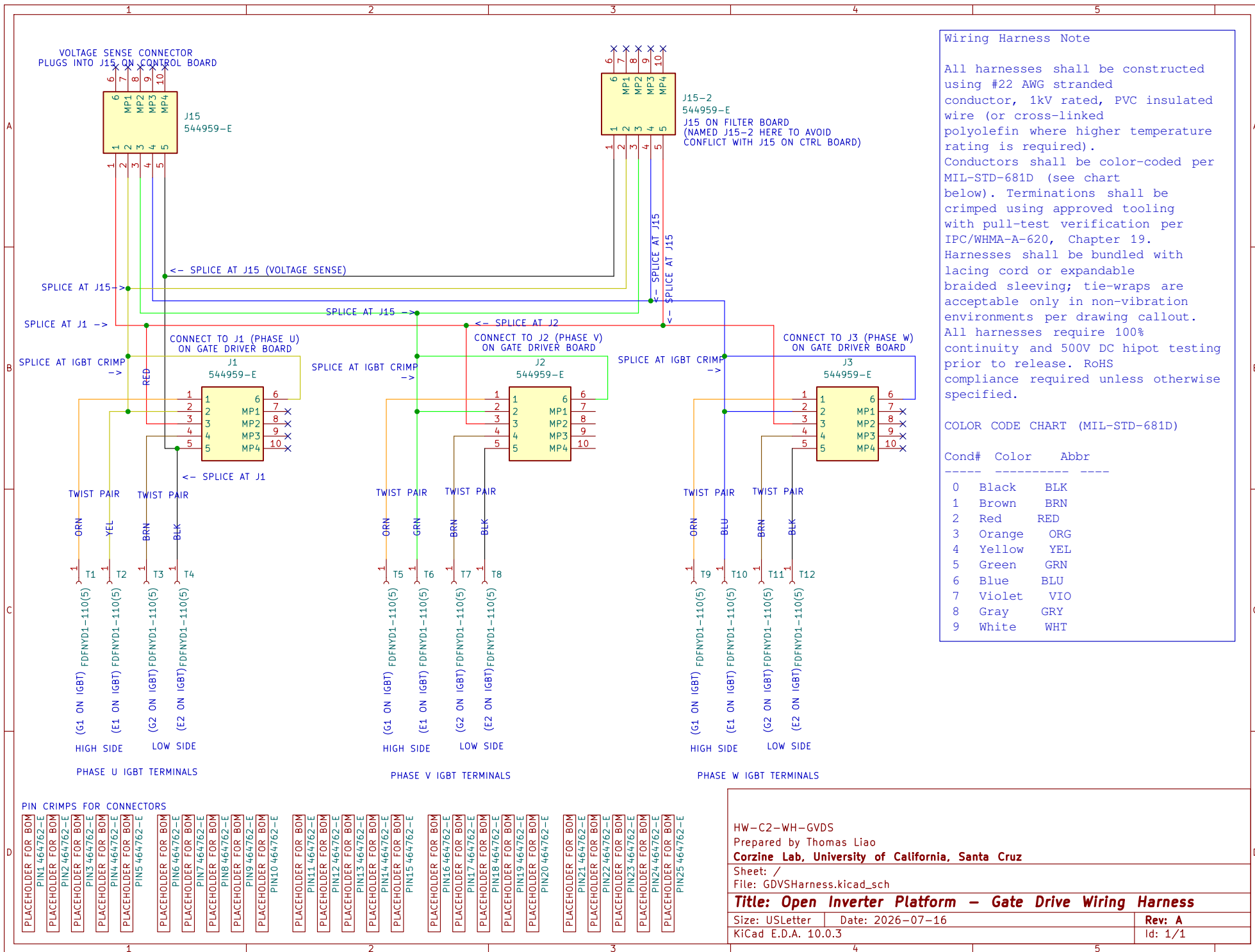
CURRENTSENSEHARNESS

HW-C2-WH-CS-A · wiring harness — schematic



GDVSHARNESS

HW-C2-WH-GDVS-A · wiring harness — schematic



TEMPSENSEHARNESS

HW-C2-WH-TSIGBT-A · wiring harness — schematic

